

Home Credit Default Risk

Exploratory Data Analysis Report

Phase 1 (application_train) & Phase 3 (Joined Dataset)

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Section 1 — Data Structure Summary (Phase 1)

Dataset shape: **307,511 rows × 122 columns**. Duplicate SK_ID_CURR: **0**.

1.1 Column Counts by dtype

dtype	count
numeric (int64/float64)	106
categorical (object)	16

1.2 Pseudo-Categorical Numeric Columns (values only in {0,1,2})

column
TARGET
FLAG_MOBIL
FLAG_EMP_PHONE
FLAG_WORK_PHONE
FLAG_CONT_MOBILE
FLAG_PHONE
FLAG_EMAIL
REG_REGION_NOT_LIVE_REGION
REG_REGION_NOT_WORK_REGION
LIVE_REGION_NOT_WORK_REGION
REG_CITY_NOT_LIVE_CITY
REG_CITY_NOT_WORK_CITY
LIVE_CITY_NOT_WORK_CITY
FLAG_DOCUMENT_2
FLAG_DOCUMENT_3
FLAG_DOCUMENT_4
FLAG_DOCUMENT_5
FLAG_DOCUMENT_6
FLAG_DOCUMENT_7
FLAG_DOCUMENT_8
FLAG_DOCUMENT_9
FLAG_DOCUMENT_10
FLAG_DOCUMENT_11
FLAG_DOCUMENT_12
FLAG_DOCUMENT_13
FLAG_DOCUMENT_14
FLAG_DOCUMENT_15
FLAG_DOCUMENT_16
FLAG_DOCUMENT_17
FLAG_DOCUMENT_18
FLAG_DOCUMENT_19
FLAG_DOCUMENT_20

column
FLAG_DOCUMENT_21

1.3 Null Rate by Column (all columns, sorted descending)

column	null_count	null_pct	flag
COMMONAREA_AVG	214865	69.87	>60% DROP
COMMONAREA_MODE	214865	69.87	>60% DROP
COMMONAREA_MEDI	214865	69.87	>60% DROP
NONLIVINGAPARTMENTS_MEDI	213514	69.43	>60% DROP
NONLIVINGAPARTMENTS_MODE	213514	69.43	>60% DROP
NONLIVINGAPARTMENTS_AVG	213514	69.43	>60% DROP
FONDKAPREMONT_MODE	210295	68.39	>60% DROP
LIVINGAPARTMENTS_AVG	210199	68.35	>60% DROP
LIVINGAPARTMENTS_MEDI	210199	68.35	>60% DROP
LIVINGAPARTMENTS_MODE	210199	68.35	>60% DROP
FLOORSMIN_MODE	208642	67.85	>60% DROP
FLOORSMIN_AVG	208642	67.85	>60% DROP
FLOORSMIN_MEDI	208642	67.85	>60% DROP
YEARS_BUILD_AVG	204488	66.5	>60% DROP
YEARS_BUILD_MODE	204488	66.5	>60% DROP
YEARS_BUILD_MEDI	204488	66.5	>60% DROP
OWN_CAR_AGE	202929	65.99	>60% DROP
LANDAREA_MEDI	182590	59.38	1-60% impute
LANDAREA_AVG	182590	59.38	1-60% impute
LANDAREA_MODE	182590	59.38	1-60% impute
BASEMENTAREA_MODE	179943	58.52	1-60% impute
BASEMENTAREA_MEDI	179943	58.52	1-60% impute
BASEMENTAREA_AVG	179943	58.52	1-60% impute
EXT_SOURCE_1	173378	56.38	1-60% impute
NONLIVINGAREA_MODE	169682	55.18	1-60% impute
NONLIVINGAREA_AVG	169682	55.18	1-60% impute
NONLIVINGAREA_MEDI	169682	55.18	1-60% impute
ELEVATORS_AVG	163891	53.3	1-60% impute
ELEVATORS_MEDI	163891	53.3	1-60% impute
ELEVATORS_MODE	163891	53.3	1-60% impute
WALLSMATERIAL_MODE	156341	50.84	1-60% impute
APARTMENTS_AVG	156061	50.75	1-60% impute
APARTMENTS_MEDI	156061	50.75	1-60% impute
APARTMENTS_MODE	156061	50.75	1-60% impute
ENTRANCES_MODE	154828	50.35	1-60% impute
ENTRANCES_MEDI	154828	50.35	1-60% impute
ENTRANCES_AVG	154828	50.35	1-60% impute
LIVINGAREA_AVG	154350	50.19	1-60% impute
LIVINGAREA_MEDI	154350	50.19	1-60% impute
LIVINGAREA_MODE	154350	50.19	1-60% impute
HOUSETYPE_MODE	154297	50.18	1-60% impute

column	null_count	null_pct	flag
FLOORSMAX_MODE	153020	49.76	1-60% impute
FLOORSMAX_AVG	153020	49.76	1-60% impute
FLOORSMAX_MEDI	153020	49.76	1-60% impute
YEARS_BEGINEXPLUATATION_MODE	150007	48.78	1-60% impute
YEARS_BEGINEXPLUATATION_MEDI	150007	48.78	1-60% impute
YEARS_BEGINEXPLUATATION_AVG	150007	48.78	1-60% impute
TOTALAREA_MODE	148431	48.27	1-60% impute
EMERGENCYSTATE_MODE	145755	47.4	1-60% impute
OCCUPATION_TYPE	96391	31.35	1-60% impute
EXT_SOURCE_3	60965	19.83	1-60% impute
ORGANIZATION_TYPE	55374	18.01	1-60% impute
DAYS_EMPLOYED	55374	18.01	1-60% impute
AMT_REQ_CREDIT_BUREAU_WEEK	41519	13.5	1-60% impute
AMT_REQ_CREDIT_BUREAU_HOUR	41519	13.5	1-60% impute
AMT_REQ_CREDIT_BUREAU_DAY	41519	13.5	1-60% impute
AMT_REQ_CREDIT_BUREAU_YEAR	41519	13.5	1-60% impute
AMT_REQ_CREDIT_BUREAU_QRT	41519	13.5	1-60% impute
AMT_REQ_CREDIT_BUREAU_MON	41519	13.5	1-60% impute
NAME_TYPE_SUITE	1292	0.42	1-60% impute
DEF_60_CNT_SOCIAL_CIRCLE	1021	0.33	1-60% impute
OBS_30_CNT_SOCIAL_CIRCLE	1021	0.33	1-60% impute
DEF_30_CNT_SOCIAL_CIRCLE	1021	0.33	1-60% impute
OBS_60_CNT_SOCIAL_CIRCLE	1021	0.33	1-60% impute
EXT_SOURCE_2	660	0.21	1-60% impute
AMT_GOODS_PRICE	278	0.09	1-60% impute
HOUR_APPR_PROCESS_START	0	0	0% clean
WEEKDAY_APPR_PROCESS_START	0	0	0% clean
REG_REGION_NOT_LIVE_REGION	0	0	0% clean
REG_REGION_NOT_WORK_REGION	0	0	0% clean
REG_CITY_NOT_WORK_CITY	0	0	0% clean
CODE_GENDER	4	0	0% clean
SK_ID_CURR	0	0	0% clean
TARGET	0	0	0% clean
AMT_ANNUITY	12	0	0% clean
AMT_CREDIT	0	0	0% clean
AMT_INCOME_TOTAL	0	0	0% clean
CNT_CHILDREN	0	0	0% clean
NAME_FAMILY_STATUS	0	0	0% clean
NAME_HOUSING_TYPE	0	0	0% clean
NAME_INCOME_TYPE	0	0	0% clean
NAME_EDUCATION_TYPE	0	0	0% clean
DAYS_BIRTH	0	0	0% clean
REGION_POPULATION_RELATIVE	0	0	0% clean
DAYS_REGISTRATION	0	0	0% clean
DAYS_ID_PUBLISH	0	0	0% clean

column	null_count	null_pct	flag
FLAG_MOBIL	0	0	0% clean
FLAG_EMP_PHONE	0	0	0% clean
FLAG_WORK_PHONE	0	0	0% clean
FLAG_CONT_MOBILE	0	0	0% clean
FLAG_OWN_REALTY	0	0	0% clean
FLAG_OWN_CAR	0	0	0% clean
NAME_CONTRACT_TYPE	0	0	0% clean
FLAG_PHONE	0	0	0% clean
REGION_RATING_CLIENT	0	0	0% clean
REGION_RATING_CLIENT_W_CITY	0	0	0% clean
FLAG_EMAIL	0	0	0% clean
CNT_FAM_MEMBERS	2	0	0% clean
REG_CITY_NOT_LIVE_CITY	0	0	0% clean
LIVE_REGION_NOT_WORK_REGION	0	0	0% clean
LIVE_CITY_NOT_WORK_CITY	0	0	0% clean
FLAG_DOCUMENT_5	0	0	0% clean
FLAG_DOCUMENT_4	0	0	0% clean
DAYS_LAST_PHONE_CHANGE	1	0	0% clean
FLAG_DOCUMENT_2	0	0	0% clean
FLAG_DOCUMENT_3	0	0	0% clean
FLAG_DOCUMENT_11	0	0	0% clean
FLAG_DOCUMENT_10	0	0	0% clean
FLAG_DOCUMENT_9	0	0	0% clean
FLAG_DOCUMENT_8	0	0	0% clean
FLAG_DOCUMENT_7	0	0	0% clean
FLAG_DOCUMENT_6	0	0	0% clean
FLAG_DOCUMENT_12	0	0	0% clean
FLAG_DOCUMENT_13	0	0	0% clean
FLAG_DOCUMENT_19	0	0	0% clean
FLAG_DOCUMENT_18	0	0	0% clean
FLAG_DOCUMENT_17	0	0	0% clean
FLAG_DOCUMENT_16	0	0	0% clean
FLAG_DOCUMENT_15	0	0	0% clean
FLAG_DOCUMENT_14	0	0	0% clean
FLAG_DOCUMENT_20	0	0	0% clean
FLAG_DOCUMENT_21	0	0	0% clean

Columns with >60% nulls are drop candidates. Null rates computed after replacing 365243/XNA/XAP.

1.4 Numeric Column Min / Max

column	min	max	flag
SK_ID_CURR	1e+05	4.563e+05	
TARGET	0	1	
CNT_CHILDREN	0	19	
AMT_INCOME_TOTAL	2.565e+04	1.17e+08	

column	min	max	flag
AMT_CREDIT	4.5e+04	4.05e+06	
AMT_ANNUITY	1616	2.58e+05	
AMT_GOODS_PRICE	4.05e+04	4.05e+06	
REGION_POPULATION_RELATIVE	0.0003	0.0725	
DAYS_BIRTH	7489	2.523e+04	
DAYS_EMPLOYED	0	1.791e+04	
DAYS_REGISTRATION	0	2.467e+04	
DAYS_ID_PUBLISH	0	7197	
OWN_CAR_AGE	0	91	
FLAG_MOBIL	0	1	
FLAG_EMP_PHONE	0	1	
FLAG_WORK_PHONE	0	1	
FLAG_CONT_MOBILE	0	1	
FLAG_PHONE	0	1	
FLAG_EMAIL	0	1	
CNT_FAM_MEMBERS	1	20	
REGION_RATING_CLIENT	1	3	
REGION_RATING_CLIENT_W_CITY	1	3	
HOUR_APPR_PROCESS_START	0	23	
REG_REGION_NOT_LIVE_REGION	0	1	
REG_REGION_NOT_WORK_REGION	0	1	
LIVE_REGION_NOT_WORK_REGION	0	1	
REG_CITY_NOT_LIVE_CITY	0	1	
REG_CITY_NOT_WORK_CITY	0	1	
LIVE_CITY_NOT_WORK_CITY	0	1	
EXT_SOURCE_1	0.0146	0.9627	
EXT_SOURCE_2	0	0.855	
EXT_SOURCE_3	0.0005	0.896	
APARTMENTS_AVG	0	1	
BASEMENTAREA_AVG	0	1	
YEARS_BEGINEXPLUATATION_AVG	0	1	
YEARS_BUILD_AVG	0	1	
COMMONAREA_AVG	0	1	
ELEVATORS_AVG	0	1	
ENTRANCES_AVG	0	1	
FLOORSMAX_AVG	0	1	
FLOORSMIN_AVG	0	1	
LANDAREA_AVG	0	1	
LIVINGAPARTMENTS_AVG	0	1	
LIVINGAREA_AVG	0	1	
NONLIVINGAPARTMENTS_AVG	0	1	
NONLIVINGAREA_AVG	0	1	
APARTMENTS_MODE	0	1	
BASEMENTAREA_MODE	0	1	
YEARS_BEGINEXPLUATATION_MODE	0	1	

column	min	max	flag
YEARS_BUILD_MODE	0	1	
COMMONAREA_MODE	0	1	
ELEVATORS_MODE	0	1	
ENTRANCES_MODE	0	1	
FLOORSMAX_MODE	0	1	
FLOORSMIN_MODE	0	1	
LANDAREA_MODE	0	1	
LIVINGAPARTMENTS_MODE	0	1	
LIVINGAREA_MODE	0	1	
NONLIVINGAPARTMENTS_MODE	0	1	
NONLIVINGAREA_MODE	0	1	
APARTMENTS_MEDI	0	1	
BASEMENTAREA_MEDI	0	1	
YEARS_BEGINEXPLUATATION_MEDI	0	1	
YEARS_BUILD_MEDI	0	1	
COMMONAREA_MEDI	0	1	
ELEVATORS_MEDI	0	1	
ENTRANCES_MEDI	0	1	
FLOORSMAX_MEDI	0	1	
FLOORSMIN_MEDI	0	1	
LANDAREA_MEDI	0	1	
LIVINGAPARTMENTS_MEDI	0	1	
LIVINGAREA_MEDI	0	1	
NONLIVINGAPARTMENTS_MEDI	0	1	
NONLIVINGAREA_MEDI	0	1	
TOTALAREA_MODE	0	1	
OBS_30_CNT_SOCIAL_CIRCLE	0	348	
DEF_30_CNT_SOCIAL_CIRCLE	0	34	
OBS_60_CNT_SOCIAL_CIRCLE	0	344	
DEF_60_CNT_SOCIAL_CIRCLE	0	24	
DAYS_LAST_PHONE_CHANGE	0	4292	
FLAG_DOCUMENT_2	0	1	
FLAG_DOCUMENT_3	0	1	
FLAG_DOCUMENT_4	0	1	
FLAG_DOCUMENT_5	0	1	
FLAG_DOCUMENT_6	0	1	
FLAG_DOCUMENT_7	0	1	
FLAG_DOCUMENT_8	0	1	
FLAG_DOCUMENT_9	0	1	
FLAG_DOCUMENT_10	0	1	
FLAG_DOCUMENT_11	0	1	
FLAG_DOCUMENT_12	0	1	
FLAG_DOCUMENT_13	0	1	
FLAG_DOCUMENT_14	0	1	
FLAG_DOCUMENT_15	0	1	

column	min	max	flag
FLAG_DOCUMENT_16	0	1	
FLAG_DOCUMENT_17	0	1	
FLAG_DOCUMENT_18	0	1	
FLAG_DOCUMENT_19	0	1	
FLAG_DOCUMENT_20	0	1	
FLAG_DOCUMENT_21	0	1	
AMT_REQ_CREDIT_BUREAU_HOUR	0	4	
AMT_REQ_CREDIT_BUREAU_DAY	0	9	
AMT_REQ_CREDIT_BUREAU_WEEK	0	8	
AMT_REQ_CREDIT_BUREAU_MON	0	27	
AMT_REQ_CREDIT_BUREAU_QRT	0	261	
AMT_REQ_CREDIT_BUREAU_YEAR	0	25	

Flag column marks implausible values (e.g. negative amounts, age outside expected range).

1.5 Categorical Unique Value Counts

column	n_unique	high_cardinality
ORGANIZATION_TYPE	57	True
OCCUPATION_TYPE	18	False
NAME_INCOME_TYPE	8	False
NAME_TYPE_SUITE	7	False
WALLSMATERIAL_MODE	7	False
WEEKDAY_APPR_PROCESS_START	7	False
NAME_FAMILY_STATUS	6	False
NAME_HOUSING_TYPE	6	False
NAME_EDUCATION_TYPE	5	False
FONDKAPREMONT_MODE	4	False
HOUSETYPE_MODE	3	False
FLAG_OWN_REALTY	2	False
NAME_CONTRACT_TYPE	2	False
CODE_GENDER	2	False
FLAG_OWN_CAR	2	False
EMERGENCYSTATE_MODE	2	False

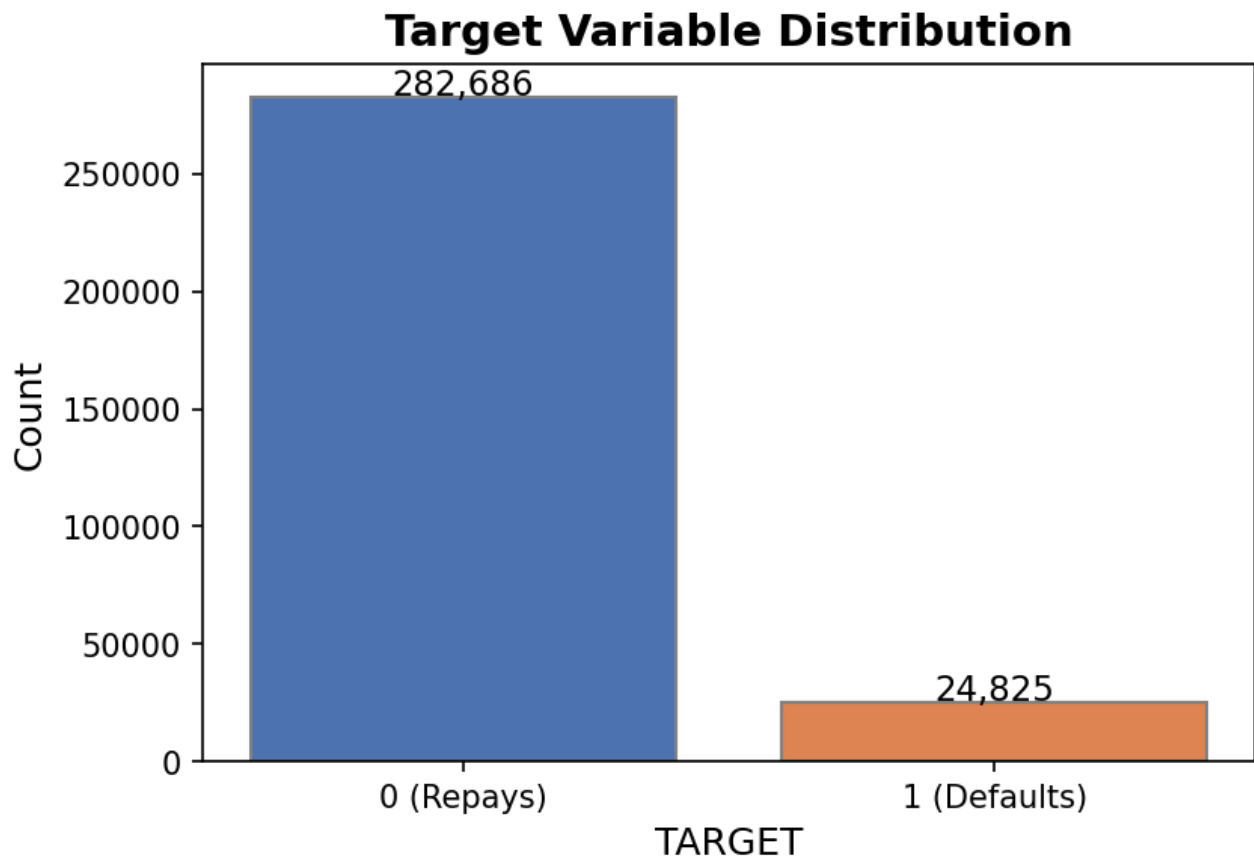
Columns with >20 unique values flagged as high cardinality — require special encoding.

Section 2 — Target Variable (Phase 1)

TARGET	count	pct
0 (Repay)	282686	91.93
1 (Defaults)	24825	8.07

Positive rate (default rate): **0.0807** (8.07%)

scale_pos_weight for XGBoost: **11.39** (= count(0) / count(1))



Class distribution. ~92% applicants repay (TARGET=0), ~8% default (TARGET=1). Strong class imbalance — use scale_pos_weight or stratified sampling.

Section 3 — Numeric Feature Signal (Phase 1)

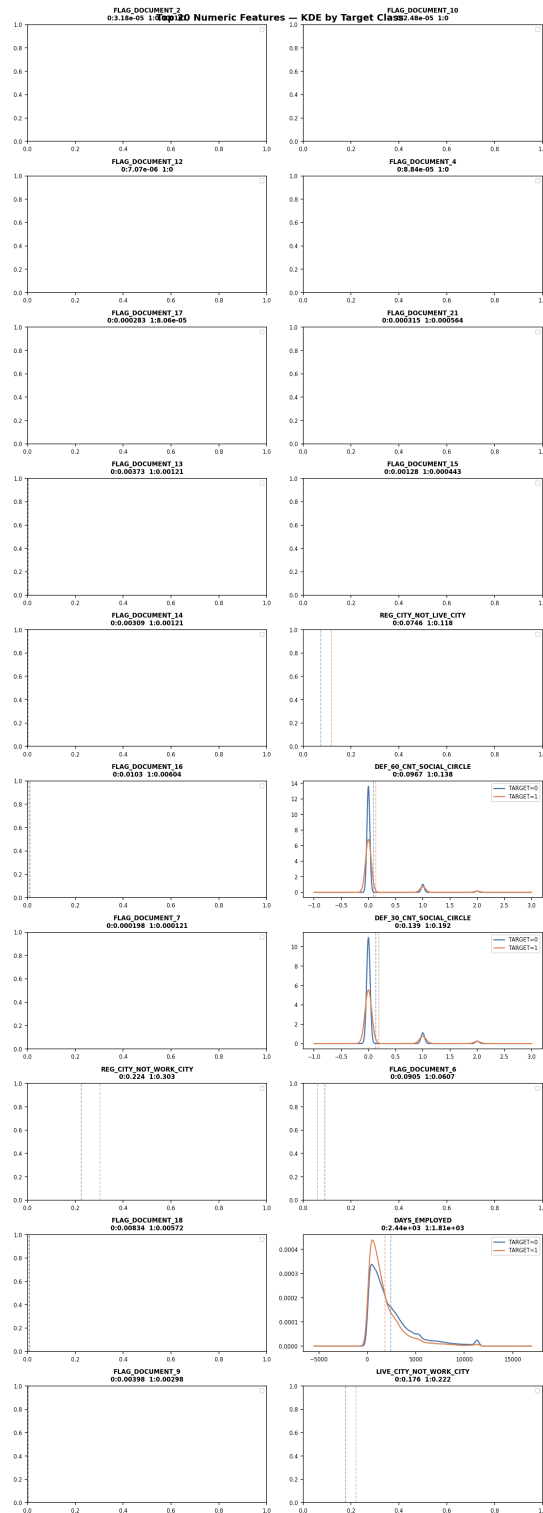
3.1 Top 30 Features by Normalised Mean Difference Between Target Classes

Normalised difference = $|\text{mean}(\text{TARGET}=1) - \text{mean}(\text{TARGET}=0)| / |\text{overall mean}|$. Higher values indicate stronger separation.

feature	mean_target0	mean_target1	norm_diff
FLAG_DOCUMENT_2	0	0.0002	3.058
FLAG_DOCUMENT_10	0	0	1.088
FLAG_DOCUMENT_12	0	0	1.088
FLAG_DOCUMENT_4	0.0001	0	1.088
FLAG_DOCUMENT_17	0.0003	0.0001	0.7592
FLAG_DOCUMENT_21	0.0003	0.0006	0.7437
FLAG_DOCUMENT_13	0.0037	0.0012	0.7149
FLAG_DOCUMENT_15	0.0013	0.0004	0.6894
FLAG_DOCUMENT_14	0.0031	0.0012	0.6401
REG_CITY_NOT_LIVE_CITY	0.0746	0.1184	0.5596
FLAG_DOCUMENT_16	0.0103	0.006	0.4258
DEF_60_CNT_SOCIAL_CIRCLE	0.0967	0.1382	0.4154
FLAG_DOCUMENT_7	0.0002	0.0001	0.4027
DEF_30_CNT_SOCIAL_CIRCLE	0.1391	0.192	0.3684
REG_CITY_NOT_WORK_CITY	0.2241	0.3029	0.3421
FLAG_DOCUMENT_6	0.0905	0.0607	0.3379
FLAG_DOCUMENT_18	0.0083	0.0057	0.3224
DAYS_EMPLOYED	2438	1815	0.2614
FLAG_DOCUMENT_9	0.004	0.003	0.2555
LIVE_CITY_NOT_WORK_CITY	0.1759	0.2217	0.2552
EXT_SOURCE_3	0.521	0.3907	0.255
EXT_SOURCE_1	0.5115	0.387	0.2479
FLAG_DOCUMENT_11	0.004	0.003	0.2477
ELEVATORS_MEDI	0.0793	0.0614	0.2301
ELEVATORS_AVG	0.0802	0.062	0.23
ELEVATORS_MODE	0.0756	0.0589	0.2251
EXT_SOURCE_2	0.5235	0.4109	0.2188
FLAG_WORK_PHONE	0.196	0.2378	0.2098
FLAG_DOCUMENT_19	0.0006	0.0005	0.2042
DAYS_LAST_PHONE_CHANGE	976.4	808.8	0.1741

Top 30 numeric features by normalised class mean difference.

3.2 KDE Plots — Top 20 Features by Target Class



Kernel density plots for the top 20 numeric features. Blue = TARGET=0 (repays), orange = TARGET=1 (defaults). Dashed vertical lines show class means. Wide separation indicates high predictive value.

3.3 Highly Skewed Features ($|\text{skewness}| > 2$)

feature	skewness	min	max	mean	p99
FLAG_MOBIL	-554.5	0	1	1	1
FLAG_DOCUMENT_12	392.1	0	1	0	0

feature	skewness	min	max	mean	p99
AMT_INCOME_TOTAL	391.6	2.565e+04	1.17e+08	1.688e+05	4.725e+05
FLAG_DOCUMENT_10	209.6	0	1	0	0
FLAG_DOCUMENT_2	153.8	0	1	0	0
AMT_REQ_CREDIT_BUREAU_QRT	134.4	0	261	0.2655	2
FLAG_DOCUMENT_4	110.9	0	1	0.0001	0
FLAG_DOCUMENT_7	72.17	0	1	0.0002	0
FLAG_DOCUMENT_17	61.21	0	1	0.0003	0
FLAG_DOCUMENT_21	54.61	0	1	0.0003	0
FLAG_DOCUMENT_20	44.37	0	1	0.0005	0
FLAG_DOCUMENT_19	40.96	0	1	0.0006	0
FLAG_DOCUMENT_15	28.7	0	1	0.0012	0
AMT_REQ_CREDIT_BUREAU_DAY	27.04	0	9	0.007	0
FLAG_CONT_MOBILE	-23.08	0	1	0.9981	1
FLAG_DOCUMENT_14	18.37	0	1	0.0029	0
FLAG_DOCUMENT_13	16.75	0	1	0.0035	0
NONLIVINGAPARTMENTS_MODE	16.25	0	1	0.0081	0.0973
FLAG_DOCUMENT_9	15.93	0	1	0.0039	0
FLAG_DOCUMENT_11	15.89	0	1	0.0039	0
NONLIVINGAPARTMENTS_MEDI	15.67	0	1	0.0087	0.1048
YEARS_BEGINEXPLUATATION_MEDI	-15.57	0	1	0.9778	0.999
NONLIVINGAPARTMENTS_AVG	15.54	0	1	0.0088	0.1081
YEARS_BEGINEXPLUATATION_AVG	-15.52	0	1	0.9777	0.999
YEARS_BEGINEXPLUATATION_MODE	-14.76	0	1	0.9771	0.999
AMT_REQ_CREDIT_BUREAU_HOUR	14.53	0	4	0.0064	0
OBS_30_CNT_SOCIAL_CIRCLE	12.14	0	348	1.422	10
OBS_60_CNT_SOCIAL_CIRCLE	12.07	0	344	1.405	10
FLAG_DOCUMENT_18	10.96	0	1	0.0081	0
FLAG_DOCUMENT_16	9.886	0	1	0.0099	0
AMT_REQ_CREDIT_BUREAU_WEEK	9.294	0	8	0.0344	1
FLAG_DOCUMENT_5	7.948	0	1	0.0151	1
REG_REGION_NOT_LIVE_REGION	7.94	0	1	0.0151	1
AMT_REQ_CREDIT_BUREAU_MONTH	7.805	0	27	0.2674	4
NONLIVINGAREA_AVG	6.559	0	1	0.0284	0.3165
NONLIVINGAREA_MODE	6.522	0	1	0.027	0.3221
NONLIVINGAREA_MEDI	6.509	0	1	0.0282	0.3213
COMMONAREA_MODE	5.621	0	1	0.0426	0.3656
COMMONAREA_AVG	5.457	0	1	0.0446	0.3766
COMMONAREA_MEDI	5.419	0	1	0.0446	0.3786
DEF_60_CNT_SOCIAL_CIRCLE	5.278	0	24	0.1	2

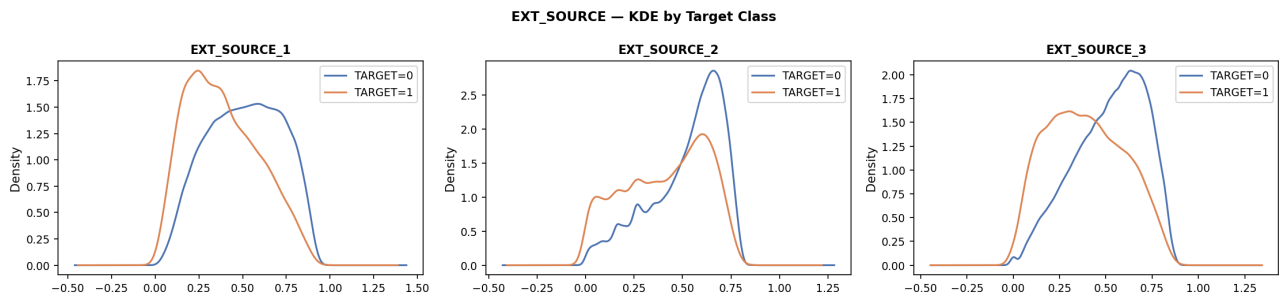
feature	skewness	min	max	mean	p99
DEF_30_CNT_SOCIAL_CIRCLE	5.184	0	34	0.1434	2
LIVE_REGION_NOT_WORK_REGION	4.652	0	1	0.0407	1
LANDAREA_AVG	4.459	0	1	0.0663	0.3746
LANDAREA_MODE	4.377	0	1	0.065	0.3789
LANDAREA_MEDI	4.368	0	1	0.0672	0.3813
REG_REGION_NOT_WORK_REGION	4.093	0	1	0.0508	1
FLAG_EMAIL	3.833	0	1	0.0567	1
BASEMENTAREA_AVG	3.566	0	1	0.0884	0.386
BASEMENTAREA_MEDI	3.553	0	1	0.088	0.3845
BASEMENTAREA_MODE	3.482	0	1	0.0875	0.3962
REG_CITY_NOT_LIVE_CITY	3.143	0	1	0.0782	1
FLAG_DOCUMENT_8	3.062	0	1	0.0814	1
LIVINGAPARTMENTS_AVG	3.042	0	1	0.1008	0.4539
LIVINGAPARTMENTS_MEDI	2.988	0	1	0.102	0.4617
FLAG_DOCUMENT_6	2.907	0	1	0.0881	1
LIVINGAPARTMENTS_MODE	2.903	0	1	0.1056	0.4885
LIVINGAREA_MODE	2.902	0	1	0.106	0.5616
LIVINGAREA_AVG	2.855	0	1	0.1074	0.5509
LIVINGAREA_MEDI	2.849	0	1	0.1086	0.5581
TOTALAREA_MODE	2.798	0	1	0.1025	0.5258
OWN_CAR_AGE	2.745	0	91	12.06	64
APARTMENTS_MODE	2.703	0	1	0.1142	0.541
APARTMENTS_AVG	2.642	0	1	0.1174	0.5449
APARTMENTS_MEDI	2.639	0	1	0.1178	0.5501
ELEVATORS_MODE	2.552	0	1	0.0745	0.6042
ELEVATORS_MEDI	2.458	0	1	0.0781	0.6
ELEVATORS_AVG	2.439	0	1	0.0789	0.6
ENTRANCES_AVG	2.4	0	1	0.1497	0.5172
ENTRANCES_MODE	2.392	0	1	0.1452	0.5172
ENTRANCES_MEDI	2.388	0	1	0.1492	0.5172

Features with absolute skewness > 2 are candidates for log transformation before modeling.

3.4 EXT_SOURCE Columns — Priority Signal Features

column	null_rate	mean_target0	mean_target1
EXT_SOURCE_1	0.5638	0.5115	0.387
EXT_SOURCE_2	0.0021	0.5235	0.4109
EXT_SOURCE_3	0.1983	0.521	0.3907
feature	EXT_SOURCE_1	EXT_SOURCE_2	EXT_SOURCE_3
EXT_SOURCE_1	1	0.214	0.1868
EXT_SOURCE_2	0.214	1	0.1092
EXT_SOURCE_3	0.1868	0.1092	1

Pairwise correlation among EXT_SOURCE columns.



EXT_SOURCE KDE plots by target class. These external credit scores are typically among the strongest predictors of default.

3.5 FLAG Columns — Default Rate Analysis

flag	default_rate_0	default_rate_1	abs_diff
FLAG_DOCUMENT_2	0.0807	0.3077	0.227
FLAG_MOBIL	0	0.0807	0.0807
FLAG_DOCUMENT_4	0.0807	0	0.0807
FLAG_DOCUMENT_12	0.0807	0	0.0807
FLAG_DOCUMENT_10	0.0807	0	0.0807
FLAG_DOCUMENT_17	0.0807	0.0244	0.0564
FLAG_DOCUMENT_21	0.0807	0.1359	0.0552
FLAG_DOCUMENT_13	0.0809	0.0277	0.0532
FLAG_DOCUMENT_15	0.0808	0.0296	0.0512
FLAG_DOCUMENT_14	0.0809	0.0332	0.0476
FLAG_EMP_PHONE	0.054	0.0866	0.0326
FLAG_DOCUMENT_16	0.081	0.0491	0.0319
FLAG_DOCUMENT_7	0.0807	0.0508	0.0299
FLAG_DOCUMENT_6	0.0831	0.0557	0.0275
FLAG_DOCUMENT_3	0.0618	0.0884	0.0266
FLAG_DOCUMENT_18	0.0809	0.0568	0.0241
FLAG_WORK_PHONE	0.0769	0.0963	0.0194
FLAG_DOCUMENT_9	0.0808	0.0618	0.019
FLAG_DOCUMENT_11	0.0808	0.0623	0.0185
FLAG_DOCUMENT_19	0.0807	0.0656	0.0152
FLAG_PHONE	0.0848	0.0704	0.0144
FLAG_DOCUMENT_8	0.0814	0.0734	0.008
FLAG_DOCUMENT_20	0.0807	0.0833	0.0026
FLAG_CONT_MOBILE	0.0784	0.0807	0.0023
FLAG_EMAIL	0.0808	0.0788	0.0021
FLAG_DOCUMENT_5	0.0807	0.08	0.0007

FLAG columns sorted by absolute difference in default rate between FLAG=0 and FLAG=1. High-ranked flags identify applicant characteristics associated with elevated default risk.

Section 4 — Categorical Feature Signal (Phase 1)

4.x NAME_CONTRACT_TYPE (2 unique values)

category	count	default_rate	anomaly
Cash loans	278232	0.0835	
Revolving loans	29279	0.0548	

NAME_CONTRACT_TYPE: sorted by default rate descending. "HIGH >50%" flags categories with default rate >50% above base rate (0.08). "LOW <50%" flags those >50% below.

4.x CODE_GENDER (3 unique values)

category	count	default_rate	anomaly
M	105059	0.1014	
F	202448	0.07	
N/A	4	0	LOW <50%

CODE_GENDER: sorted by default rate descending. "HIGH >50%" flags categories with default rate >50% above base rate (0.08). "LOW <50%" flags those >50% below.

4.x FLAG_OWN_CAR (2 unique values)

category	count	default_rate	anomaly
N	202924	0.085	
Y	104587	0.0724	

FLAG_OWN_CAR: sorted by default rate descending. "HIGH >50%" flags categories with default rate >50% above base rate (0.08). "LOW <50%" flags those >50% below.

4.x FLAG_OWN_REALTY (2 unique values)

category	count	default_rate	anomaly
N	94199	0.0832	
Y	213312	0.0796	

FLAG_OWN_REALTY: sorted by default rate descending. "HIGH >50%" flags categories with default rate >50% above base rate (0.08). "LOW <50%" flags those >50% below.

4.x NAME_TYPE_SUITE (8 unique values)

category	count	default_rate	anomaly
Other_B	1770	0.0983	
Other_A	866	0.0878	
Group of people	271	0.0849	
Unaccompanied	248526	0.0818	
Spouse, partner	11370	0.0787	
Family	40149	0.0749	

category	count	default_rate	anomaly
Children	3267	0.0738	
N/A	1292	0.0542	

NAME_TYPE_SUITE: sorted by default rate descending. "HIGH >50%" flags categories with default rate >50% above base rate (0.08). "LOW <50%" flags those >50% below.

4.x NAME_INCOME_TYPE (8 unique values)

category	count	default_rate	anomaly
Maternity leave	5	0.4	HIGH >50%
Unemployed	22	0.3636	HIGH >50%
Working	158774	0.0959	
Commercial associate	71617	0.0748	
State servant	21703	0.0575	
Pensioner	55362	0.0539	
Businessman	10	0	LOW <50%
Student	18	0	LOW <50%

NAME_INCOME_TYPE: sorted by default rate descending. "HIGH >50%" flags categories with default rate >50% above base rate (0.08). "LOW <50%" flags those >50% below.

4.x NAME_EDUCATION_TYPE (5 unique values)

category	count	default_rate	anomaly
Lower secondary	3816	0.1093	
Secondary / secondary special	218391	0.0894	
Incomplete higher	10277	0.0848	
Higher education	74863	0.0536	
Academic degree	164	0.0183	LOW <50%

NAME_EDUCATION_TYPE: sorted by default rate descending. "HIGH >50%" flags categories with default rate >50% above base rate (0.08). "LOW <50%" flags those >50% below.

4.x NAME_FAMILY_STATUS (6 unique values)

category	count	default_rate	anomaly
Civil marriage	29775	0.0994	
Single / not married	45444	0.0981	
Separated	19770	0.0819	
Married	196432	0.0756	
Widow	16088	0.0582	
Unknown	2	0	LOW <50%

NAME_FAMILY_STATUS: sorted by default rate descending. "HIGH >50%" flags categories with default rate >50% above base rate (0.08). "LOW <50%" flags those >50% below.

4.x NAME_HOUSING_TYPE (6 unique values)

category	count	default_rate	anomaly
Rented apartment	4881	0.1231	HIGH >50%
With parents	14840	0.117	
Municipal apartment	11183	0.0854	
Co-op apartment	1122	0.0793	
House / apartment	272868	0.078	
Office apartment	2617	0.0657	

NAME_HOUSING_TYPE: sorted by default rate descending. "HIGH >50%" flags categories with default rate >50% above base rate (0.08). "LOW <50%" flags those >50% below.

4.x OCCUPATION_TYPE (19 unique values)

category	count	default_rate	anomaly
Low-skill Laborers	2093	0.1715	HIGH >50%
Drivers	18603	0.1133	
Waiters/barmen staff	1348	0.1128	
Security staff	6721	0.1074	
Laborers	55186	0.1058	
Cooking staff	5946	0.1044	
Sales staff	32102	0.0963	
Cleaning staff	4653	0.0961	
Realty agents	751	0.0786	
Secretaries	1305	0.0705	
Medicine staff	8537	0.067	
Private service staff	2652	0.066	
N/A	96391	0.0651	
IT staff	526	0.0646	
HR staff	563	0.0639	
Core staff	27570	0.063	
Managers	21371	0.0621	
High skill tech staff	11380	0.0616	
Accountants	9813	0.0483	

OCCUPATION_TYPE: sorted by default rate descending. "HIGH >50%" flags categories with default rate >50% above base rate (0.08). "LOW <50%" flags those >50% below.

4.x WEEKDAY_APPR_PROCESS_START (7 unique values)

category	count	default_rate	anomaly
TUESDAY	53901	0.0835	
WEDNESDAY	51934	0.0816	
FRIDAY	50338	0.0815	
THURSDAY	50591	0.081	
SUNDAY	16181	0.0793	
SATURDAY	33852	0.0789	
MONDAY	50714	0.0776	

WEEKDAY_APPR_PROCESS_START: sorted by default rate descending. "HIGH >50%" flags categories with default rate >50% above base rate (0.08). "LOW <50%" flags those >50% below.

4.x ORGANIZATION_TYPE (58 unique values)

category	count	default_rate	anomaly
Transport: type 3	1187	0.1575	HIGH >50%
Industry: type 13	67	0.1343	HIGH >50%
Industry: type 8	24	0.125	HIGH >50%
Restaurant	1811	0.1171	
Construction	6721	0.1168	
Cleaning	260	0.1115	
Industry: type 1	1039	0.1107	
Industry: type 3	3278	0.1062	
Realtor	396	0.1061	
Agriculture	2454	0.1047	
Trade: type 3	3492	0.1034	
Self-employed	38412	0.1017	
Industry: type 4	877	0.1015	
Security	3247	0.0998	
Trade: type 7	7831	0.0945	
Business Entity Type 3	67992	0.093	
Transport: type 4	5398	0.0928	
Mobile	317	0.0915	
Trade: type 1	348	0.0891	
Industry: type 11	2704	0.0865	
Business Entity Type 2	10553	0.0853	
Postal	2157	0.0844	
Advertising	429	0.0816	
Business Entity Type 1	5984	0.0814	
Industry: type 7	1307	0.0803	
Housing	2958	0.0794	
Legal Services	305	0.0787	
Transport: type 2	2204	0.078	
Other	16683	0.0764	
Telecom	577	0.0763	
Industry: type 2	458	0.0721	
Industry: type 6	112	0.0714	
Emergency	560	0.0714	
Kindergarten	6880	0.0703	
Trade: type 2	1900	0.07	
Government	10404	0.0698	
Industry: type 5	599	0.0684	
Industry: type 9	3368	0.0668	
Electricity	950	0.0663	
Services	1575	0.066	

category	count	default_rate	anomaly
Medicine	11193	0.0658	
Hotel	966	0.0642	
Industry: type 10	109	0.0642	
Trade: type 5	49	0.0612	
School	8893	0.0591	
Religion	85	0.0588	
Insurance	597	0.057	
Culture	379	0.0554	
N/A	55374	0.054	
Bank	2507	0.0519	
Military	2634	0.0513	
Police	2341	0.05	
University	1327	0.049	
Security Ministries	1974	0.0486	
Trade: type 6	631	0.046	
Transport: type 1	201	0.0448	
Industry: type 12	369	0.0379	LOW <50%
Trade: type 4	64	0.0312	LOW <50%

ORGANIZATION_TYPE: sorted by default rate descending. "HIGH >50%" flags categories with default rate >50% above base rate (0.08). "LOW <50%" flags those >50% below.

4.x FONDKAPREMONT_MODE (5 unique values)

category	count	default_rate	anomaly
N/A	210295	0.0862	
not specified	5687	0.0754	
reg oper account	73830	0.0698	
reg oper spec account	12080	0.0656	
org spec account	5619	0.0582	

FONDKAPREMONT_MODE: sorted by default rate descending. "HIGH >50%" flags categories with default rate >50% above base rate (0.08). "LOW <50%" flags those >50% below.

4.x HOUSETYPE_MODE (4 unique values)

category	count	default_rate	anomaly
specific housing	1499	0.1014	
N/A	154297	0.0915	
terraced house	1212	0.085	
block of flats	150503	0.0694	

HOUSETYPE_MODE: sorted by default rate descending. "HIGH >50%" flags categories with default rate >50% above base rate (0.08). "LOW <50%" flags those >50% below.

4.x WALLSMATERIAL_MODE (8 unique values)

category	count	default_rate	anomaly
Wooden	5362	0.097	
N/A	156341	0.0913	
Others	1625	0.0831	
Mixed	2296	0.0753	
Stone, brick	64815	0.0741	
Block	9253	0.0702	
Panel	66040	0.0635	
Monolithic	1779	0.0472	

WALLSMATERIAL_MODE: sorted by default rate descending. "HIGH >50%" flags categories with default rate >50% above base rate (0.08). "LOW <50%" flags those >50% below.

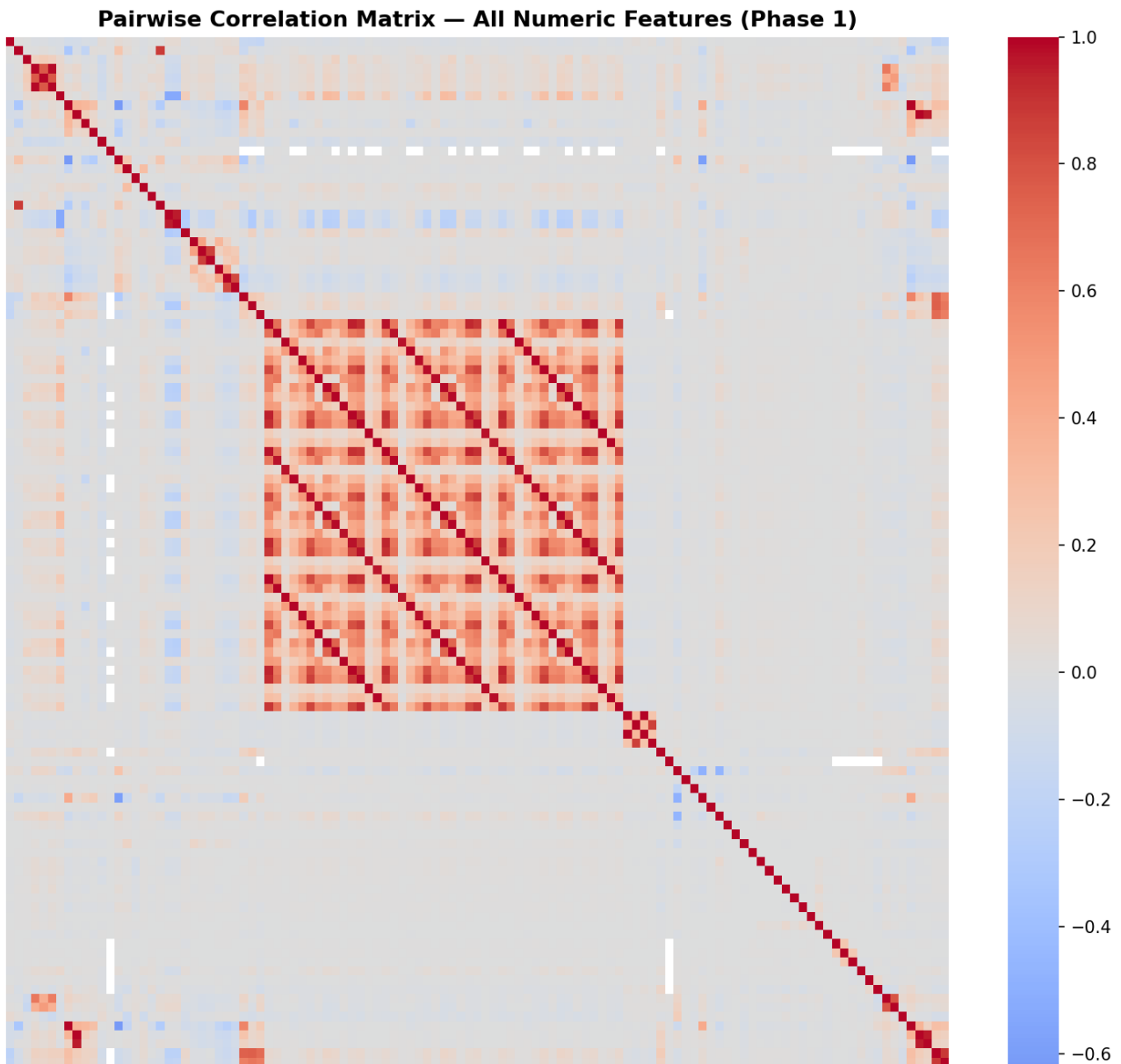
4.x EMERGENCYSTATE_MODE (3 unique values)

category	count	default_rate	anomaly
Yes	2328	0.0958	
N/A	145755	0.0926	
No	159428	0.0696	

EMERGENCYSTATE_MODE: sorted by default rate descending. "HIGH >50%" flags categories with default rate >50% above base rate (0.08). "LOW <50%" flags those >50% below.

Section 5 — Feature Correlations (Phase 1)

5.1 Full Pairwise Correlation Heatmap



Full pairwise Pearson correlation matrix for all numeric features. Red = positive correlation, blue = negative. Blocks of strong correlation indicate potential multicollinearity.

5.2 High-Correlation Pairs ($|\text{corr}| > 0.85$)

col1	col2	abs_corr	prefer
DAYS_BIRTH	AGE_YEARS	1	DAYS_BIRTH
DAYS_EMPLOYED	EMPLOYED_YEARS	1	DAYS_EMPLOYED

col1	col2	abs_corr	prefer
YEARS_BUILD_AVG	YEARS_BUILD_MEDI	0.9985	YEARS_BUILD_MEDI
OBS_30_CNT_SOCIAL_CIRCLE	OBS_60_CNT_SOCIAL_CIRCLE	0.9985	OBS_30_CNT_SOCIAL_CIRCLE
FLOORSMIN_AVG	FLOORSMIN_MEDI	0.9972	FLOORSMIN_AVG
FLOORSMAX_AVG	FLOORSMAX_MEDI	0.997	FLOORSMAX_AVG
ENTRANCES_AVG	ENTRANCES_MEDI	0.9969	ENTRANCES_AVG
ELEVATORS_AVG	ELEVATORS_MEDI	0.9961	ELEVATORS_AVG
COMMONAREA_AVG	COMMONAREA_MEDI	0.996	COMMONAREA_MEDI
LIVINGAREA_AVG	LIVINGAREA_MEDI	0.9956	LIVINGAREA_AVG
APARTMENTS_AVG	APARTMENTS_MEDI	0.9951	APARTMENTS_AVG
BASEMENTAREA_AVG	BASEMENTAREA_MEDI	0.9943	BASEMENTAREA_AVG
YEARS_BEGINEXPLUATATION_AVG	YEARS_BEGINEXPLUATATION_MEDI	0.9938	YEARS_BEGINEXPLUATATION_MEDI
LIVINGAPARTMENTS_AVG	LIVINGAPARTMENTS_MEDI	0.9938	LIVINGAPARTMENTS_AVG
LANDAREA_AVG	LANDAREA_MEDI	0.9916	LANDAREA_MEDI
NONLIVINGAPARTMENTS_AVG	NONLIVINGAPARTMENTS_MEDI	0.9908	NONLIVINGAPARTMENTS_AVG
NONLIVINGAREA_AVG	NONLIVINGAREA_MEDI	0.9904	NONLIVINGAREA_AVG
YEARS_BUILD_MODE	YEARS_BUILD_MEDI	0.9895	YEARS_BUILD_MEDI
YEARS_BUILD_AVG	YEARS_BUILD_MODE	0.9894	YEARS_BUILD_AVG
FLOORSMIN_MODE	FLOORSMIN_MEDI	0.9884	FLOORSMIN_MEDI
FLOORSMAX_MODE	FLOORSMAX_MEDI	0.9882	FLOORSMAX_MEDI
AMT_CREDIT	AMT_GOODS_PRICE	0.987	AMT_GOODS_PRICE
FLOORSMIN_AVG	FLOORSMIN_MODE	0.9859	FLOORSMIN_AVG
FLOORSMAX_AVG	FLOORSMAX_MODE	0.9857	FLOORSMAX_AVG
ELEVATORS_MODE	ELEVATORS_MEDI	0.9828	ELEVATORS_MEDI
LANDAREA_MODE	LANDAREA_MEDI	0.9808	LANDAREA_MEDI
ENTRANCES_MODE	ENTRANCES_MEDI	0.9807	ENTRANCES_MEDI
COMMONAREA_MODE	COMMONAREA_MEDI	0.9799	COMMONAREA_MEDI
ELEVATORS_AVG	ELEVATORS_MODE	0.9788	ELEVATORS_AVG
NONLIVINGAPARTMENTS_MODE	NONLIVINGAPARTMENTS_MEDI	0.9786	NONLIVINGAPARTMENTS_MEDI
BASEMENTAREA_MODE	BASEMENTAREA_MEDI	0.9779	BASEMENTAREA_MEDI
ENTRANCES_AVG	ENTRANCES_MODE	0.9777	ENTRANCES_AVG
APARTMENTS_MODE	APARTMENTS_MEDI	0.9772	APARTMENTS_MEDI
COMMONAREA_AVG	COMMONAREA_MODE	0.9771	COMMONAREA_AVG
NONLIVINGAREA_MODE	NONLIVINGAREA_MEDI	0.9758	NONLIVINGAREA_MEDI
LIVINGAPARTMENTS_MODE	LIVINGAPARTMENTS_MEDI	0.9756	LIVINGAPARTMENTS_MEDI
LIVINGAREA_MODE	LIVINGAREA_MEDI	0.9747	LIVINGAREA_MEDI
LANDAREA_AVG	LANDAREA_MODE	0.9737	LANDAREA_AVG
BASEMENTAREA_AVG	BASEMENTAREA_MODE	0.9735	BASEMENTAREA_AVG
APARTMENTS_AVG	APARTMENTS_MODE	0.9733	APARTMENTS_AVG
LIVINGAREA_AVG	LIVINGAREA_MODE	0.972	LIVINGAREA_AVG

col1	col2	abs_corr	prefer
YEARS_BEGINEXPLUATATION_AVG	YEARS_BEGINEXPLUATATION_MODE	0.9719	YEARS_BEGINEXPLUATATION_AVG
LIVINGAPARTMENTS_AVG	LIVINGAPARTMENTS_MODE	0.9701	LIVINGAPARTMENTS_AVG
NONLIVINGAPARTMENTS_AVG	NONLIVINGAPARTMENTS_MODE	0.9694	NONLIVINGAPARTMENTS_AVG
NONLIVINGAREA_AVG	NONLIVINGAREA_MODE	0.9661	NONLIVINGAREA_AVG
YEARS_BEGINEXPLUATATION_MODE	YEARS_BEGINEXPLUATATION_MEDI	0.9635	YEARS_BEGINEXPLUATATION_MEDI
DAYS_EMPLOYED	EMPLOYMENT_RATIO	0.9547	DAYS_EMPLOYED
EMPLOYED_YEARS	EMPLOYMENT_RATIO	0.9547	EMPLOYED_YEARS
REGION_RATING_CLIENT	REGION_RATING_CLIENT_W_CITY	0.9508	REGION_RATING_CLIENT_W_CITY
APARTMENTS_AVG	LIVINGAPARTMENTS_AVG	0.944	APARTMENTS_AVG
APARTMENTS_MEDI	LIVINGAPARTMENTS_MEDI	0.9425	APARTMENTS_MEDI
APARTMENTS_AVG	LIVINGAPARTMENTS_MEDI	0.9419	APARTMENTS_AVG
APARTMENTS_MODE	LIVINGAPARTMENTS_MODE	0.9378	APARTMENTS_MODE
LIVINGAPARTMENTS_AVG	APARTMENTS_MEDI	0.9356	APARTMENTS_MEDI
LIVINGAPARTMENTS_MODE	APARTMENTS_MEDI	0.9322	APARTMENTS_MEDI
APARTMENTS_AVG	LIVINGAPARTMENTS_MODE	0.9306	APARTMENTS_AVG
LIVINGAREA_AVG	TOTALAREA_MODE	0.925	LIVINGAREA_AVG
LIVINGAREA_MEDI	TOTALAREA_MODE	0.9194	LIVINGAREA_MEDI
APARTMENTS_MEDI	LIVINGAREA_MEDI	0.9159	LIVINGAREA_MEDI
APARTMENTS_MODE	LIVINGAPARTMENTS_MEDI	0.9145	APARTMENTS_MODE
APARTMENTS_AVG	LIVINGAREA_AVG	0.9136	LIVINGAREA_AVG
LIVINGAREA_AVG	APARTMENTS_MEDI	0.9125	LIVINGAREA_AVG
APARTMENTS_AVG	LIVINGAREA_MEDI	0.9123	LIVINGAREA_MEDI
APARTMENTS_MODE	LIVINGAREA_MODE	0.9104	LIVINGAREA_MODE
LIVINGAPARTMENTS_AVG	APARTMENTS_MODE	0.9083	APARTMENTS_MODE
LIVINGAREA_MODE	TOTALAREA_MODE	0.8992	TOTALAREA_MODE
LIVINGAREA_MODE	APARTMENTS_MEDI	0.8961	LIVINGAREA_MODE
APARTMENTS_MODE	LIVINGAREA_MEDI	0.8941	LIVINGAREA_MEDI
APARTMENTS_AVG	LIVINGAREA_MODE	0.8935	LIVINGAREA_MODE
APARTMENTS_AVG	TOTALAREA_MODE	0.8926	TOTALAREA_MODE
LIVINGAREA_AVG	APARTMENTS_MODE	0.8907	LIVINGAREA_AVG
APARTMENTS_MEDI	TOTALAREA_MODE	0.8866	TOTALAREA_MODE
LIVINGAPARTMENTS_MEDI	LIVINGAREA_MEDI	0.8847	LIVINGAREA_MEDI
LIVINGAREA_AVG	LIVINGAPARTMENTS_MEDI	0.8832	LIVINGAREA_AVG
LIVINGAPARTMENTS_AVG	LIVINGAREA_AVG	0.8808	LIVINGAREA_AVG
EXT_SOURCE_MEAN	EXT_SOURCE_MIN	0.8799	EXT_SOURCE_MEAN
CNT_CHILDREN	CNT_FAM_MEMBERS	0.8792	CNT_CHILDREN
LIVINGAPARTMENTS_MODE	LIVINGAREA_MODE	0.8785	LIVINGAREA_MODE
LIVINGAPARTMENTS_AVG	LIVINGAREA_MEDI	0.8783	LIVINGAREA_MEDI
LIVINGAPARTMENTS_MODE	LIVINGAREA_MEDI	0.8744	LIVINGAREA_MEDI
LIVINGAREA_AVG	LIVINGAPARTMENTS_MODE	0.8731	LIVINGAREA_AVG
ELEVATORS_MEDI	LIVINGAREA_MEDI	0.8683	ELEVATORS_MEDI
ELEVATORS_AVG	LIVINGAREA_AVG	0.8678	ELEVATORS_AVG

col1	col2	abs_corr	prefer
LIVINGAREA_AVG	ELEVATORS_MEDI	0.8658	ELEVATORS_MEDI
ELEVATORS_AVG	LIVINGAREA_MEDI	0.8656	ELEVATORS_AVG
APARTMENTS_MODE	TOTALAREA_MODE	0.8636	TOTALAREA_MODE
REG_REGION_NOT_WORK_REGION	LIVE_REGION_NOT_WORK_REGION	0.8606	REG_REGION_NOT_W ORK_REGION
DEF_30_CNT_SOCIAL_CIRCLE	DEF_60_CNT_SOCIAL_CIRCLE	0.8605	DEF_30_CNT_SOCIAL_ CIRCLE
LIVINGAREA_MODE	LIVINGAPARTMENTS_MEDI	0.8574	LIVINGAREA_MODE
ELEVATORS_MODE	LIVINGAREA_MODE	0.856	ELEVATORS_MODE
ELEVATORS_MODE	LIVINGAREA_MEDI	0.8558	LIVINGAREA_MEDI
LIVINGAREA_AVG	ELEVATORS_MODE	0.8526	LIVINGAREA_AVG
LIVINGAPARTMENTS_AVG	LIVINGAREA_MODE	0.8517	LIVINGAREA_MODE

Column pairs with absolute Pearson correlation > 0.85. "prefer" column indicates which has higher absolute correlation with TARGET — this is the preferred column to retain if one must be dropped.

5.3 Engineered Feature Candidates

feature	corr	outperforms_components
CREDIT_INCOME_RATIO	0.0077	False
ANNUITY_INCOME_RATIO	0.0143	True
CREDIT_GOODS_RATIO	0.0694	True
AGE_YEARS	0.0782	False
EMPLOYED_YEARS	0.075	False
EMPLOYMENT_RATIO	0.068	False
EXT_SOURCE_MEAN	0.2221	True
EXT_SOURCE_MIN	0.1853	True

For each engineered feature: Pearson correlation with TARGET and whether it outperforms all component features. Flagged as KEEP when correlation exceeds all components.

Section 6 — Phase 1 Output Summary

6.1 Dataset Shape

307,511 rows × 122 columns.

6.2 Column Count by Null Rate Bucket

null_bucket	n_columns
0% null	56
1–20% null	16
20–60%	33
>60% null	17

6.3 Positive Rate and scale_pos_weight

Positive rate: **0.0807**

scale_pos_weight: **11.39**

6.4 Top 10 Numeric Features by Class Separation

feature	mean_target0	mean_target1	norm_diff
FLAG_DOCUMENT_2	0	0.0002	3.058
FLAG_DOCUMENT_10	0	0	1.088
FLAG_DOCUMENT_12	0	0	1.088
FLAG_DOCUMENT_4	0.0001	0	1.088
FLAG_DOCUMENT_17	0.0003	0.0001	0.7592
FLAG_DOCUMENT_21	0.0003	0.0006	0.7437
FLAG_DOCUMENT_13	0.0037	0.0012	0.7149
FLAG_DOCUMENT_15	0.0013	0.0004	0.6894
FLAG_DOCUMENT_14	0.0031	0.0012	0.6401
REG_CITY_NOT_LIVE_CITY	0.0746	0.1184	0.5596

6.5 Top 5 Categorical Features by Default Rate Spread

column	spread
NAME_INCOME_TYPE	0.4
ORGANIZATION_TYPE	0.1263
OCCUPATION_TYPE	0.1232
CODE_GENDER	0.1014
NAME_FAMILY_STATUS	0.0994

6.6 All High-Correlation Pairs ($|\text{corr}| > 0.85$)

col1	col2	abs_corr	prefer
DAYS_BIRTH	AGE_YEARS	1	DAYS_BIRTH
DAYS_EMPLOYED	EMPLOYED_YEARS	1	DAYS_EMPLOYED
YEARS_BUILD_AVG	YEARS_BUILD_MEDI	0.9985	YEARS_BUILD_MEDI
OBS_30_CNT_SOCIAL_CIRCLE	OBS_60_CNT_SOCIAL_CIRCLE	0.9985	OBS_30_CNT_SOCIAL_CIRCLE
FLOORSMIN_AVG	FLOORSMIN_MEDI	0.9972	FLOORSMIN_AVG
FLOORSMAX_AVG	FLOORSMAX_MEDI	0.997	FLOORSMAX_AVG
ENTRANCES_AVG	ENTRANCES_MEDI	0.9969	ENTRANCES_AVG
ELEVATORS_AVG	ELEVATORS_MEDI	0.9961	ELEVATORS_AVG
COMMONAREA_AVG	COMMONAREA_MEDI	0.996	COMMONAREA_MEDI
LIVINGAREA_AVG	LIVINGAREA_MEDI	0.9956	LIVINGAREA_AVG
APARTMENTS_AVG	APARTMENTS_MEDI	0.9951	APARTMENTS_AVG
BASEMENTAREA_AVG	BASEMENTAREA_MEDI	0.9943	BASEMENTAREA_AVG
YEARS_BEGINEXPLUATATION_AVG	YEARS_BEGINEXPLUATATION_MEDI	0.9938	YEARS_BEGINEXPLUATATION_MEDI
LIVINGAPARTMENTS_AVG	LIVINGAPARTMENTS_MEDI	0.9938	LIVINGAPARTMENTS_AVG
LANDAREA_AVG	LANDAREA_MEDI	0.9916	LANDAREA_MEDI
NONLIVINGAPARTMENTS_AVG	NONLIVINGAPARTMENTS_MEDI	0.9908	NONLIVINGAPARTMENTS_AVG
NONLIVINGAREA_AVG	NONLIVINGAREA_MEDI	0.9904	NONLIVINGAREA_AVG
YEARS_BUILD_MODE	YEARS_BUILD_MEDI	0.9895	YEARS_BUILD_MEDI
YEARS_BUILD_AVG	YEARS_BUILD_MODE	0.9894	YEARS_BUILD_AVG
FLOORSMIN_MODE	FLOORSMIN_MEDI	0.9884	FLOORSMIN_MEDI
FLOORSMAX_MODE	FLOORSMAX_MEDI	0.9882	FLOORSMAX_MEDI
AMT_CREDIT	AMT_GOODS_PRICE	0.987	AMT_GOODS_PRICE
FLOORSMIN_AVG	FLOORSMIN_MODE	0.9859	FLOORSMIN_AVG
FLOORSMAX_AVG	FLOORSMAX_MODE	0.9857	FLOORSMAX_AVG
ELEVATORS_MODE	ELEVATORS_MEDI	0.9828	ELEVATORS_MEDI
LANDAREA_MODE	LANDAREA_MEDI	0.9808	LANDAREA_MEDI
ENTRANCES_MODE	ENTRANCES_MEDI	0.9807	ENTRANCES_MEDI
COMMONAREA_MODE	COMMONAREA_MEDI	0.9799	COMMONAREA_MEDI
ELEVATORS_AVG	ELEVATORS_MODE	0.9788	ELEVATORS_AVG
NONLIVINGAPARTMENTS_MODE	NONLIVINGAPARTMENTS_MEDI	0.9786	NONLIVINGAPARTMENTS_MEDI
BASEMENTAREA_MODE	BASEMENTAREA_MEDI	0.9779	BASEMENTAREA_MEDI
ENTRANCES_AVG	ENTRANCES_MODE	0.9777	ENTRANCES_AVG
APARTMENTS_MODE	APARTMENTS_MEDI	0.9772	APARTMENTS_MEDI
COMMONAREA_AVG	COMMONAREA_MODE	0.9771	COMMONAREA_AVG
NONLIVINGAREA_MODE	NONLIVINGAREA_MEDI	0.9758	NONLIVINGAREA_MEDI
LIVINGAPARTMENTS_MODE	LIVINGAPARTMENTS_MEDI	0.9756	LIVINGAPARTMENTS_MEDI
LIVINGAREA_MODE	LIVINGAREA_MEDI	0.9747	LIVINGAREA_MEDI
LANDAREA_AVG	LANDAREA_MODE	0.9737	LANDAREA_AVG
BASEMENTAREA_AVG	BASEMENTAREA_MODE	0.9735	BASEMENTAREA_AVG

col1	col2	abs_corr	prefer
APARTMENTS_AVG	APARTMENTS_MODE	0.9733	APARTMENTS_AVG
LIVINGAREA_AVG	LIVINGAREA_MODE	0.972	LIVINGAREA_AVG
YEARS_BEGINEXPLUATATION_AVG	YEARS_BEGINEXPLUATATION_MODE	0.9719	YEARS_BEGINEXPLUATATION_AVG
LIVINGAPARTMENTS_AVG	LIVINGAPARTMENTS_MODE	0.9701	LIVINGAPARTMENTS_AVG
NONLIVINGAPARTMENTS_AVG	NONLIVINGAPARTMENTS_MODE	0.9694	NONLIVINGAPARTMENTS_AVG
NONLIVINGAREA_AVG	NONLIVINGAREA_MODE	0.9661	NONLIVINGAREA_AVG
YEARS_BEGINEXPLUATATION_MEDI	YEARS_BEGINEXPLUATATION_MEDI	0.9635	YEARS_BEGINEXPLUATATION_MEDI
DAYS_EMPLOYED	EMPLOYMENT_RATIO	0.9547	DAYS_EMPLOYED
EMPLOYED_YEARS	EMPLOYMENT_RATIO	0.9547	EMPLOYED_YEARS
REGION_RATING_CLIENT	REGION_RATING_CLIENT_W_CITY	0.9508	REGION_RATING_CLIENT_W_CITY
APARTMENTS_AVG	LIVINGAPARTMENTS_AVG	0.944	APARTMENTS_AVG
APARTMENTS_MEDI	LIVINGAPARTMENTS_MEDI	0.9425	APARTMENTS_MEDI
APARTMENTS_AVG	LIVINGAPARTMENTS_MEDI	0.9419	APARTMENTS_AVG
APARTMENTS_MODE	LIVINGAPARTMENTS_MODE	0.9378	APARTMENTS_MODE
LIVINGAPARTMENTS_AVG	APARTMENTS_MEDI	0.9356	APARTMENTS_MEDI
LIVINGAPARTMENTS_MODE	APARTMENTS_MEDI	0.9322	APARTMENTS_MEDI
APARTMENTS_AVG	LIVINGAPARTMENTS_MODE	0.9306	APARTMENTS_AVG
LIVINGAREA_AVG	TOTALAREA_MODE	0.925	LIVINGAREA_AVG
LIVINGAREA_MEDI	TOTALAREA_MODE	0.9194	LIVINGAREA_MEDI
APARTMENTS_MEDI	LIVINGAREA_MEDI	0.9159	LIVINGAREA_MEDI
APARTMENTS_MODE	LIVINGAPARTMENTS_MEDI	0.9145	APARTMENTS_MODE
APARTMENTS_AVG	LIVINGAREA_AVG	0.9136	LIVINGAREA_AVG
LIVINGAREA_AVG	APARTMENTS_MEDI	0.9125	LIVINGAREA_AVG
APARTMENTS_AVG	LIVINGAREA_MEDI	0.9123	LIVINGAREA_MEDI
APARTMENTS_MODE	LIVINGAREA_MODE	0.9104	LIVINGAREA_MODE
LIVINGAPARTMENTS_AVG	APARTMENTS_MODE	0.9083	APARTMENTS_MODE
LIVINGAREA_MODE	TOTALAREA_MODE	0.8992	TOTALAREA_MODE
LIVINGAREA_MODE	APARTMENTS_MEDI	0.8961	LIVINGAREA_MODE
APARTMENTS_MODE	LIVINGAREA_MEDI	0.8941	LIVINGAREA_MEDI
APARTMENTS_AVG	LIVINGAREA_MODE	0.8935	LIVINGAREA_MODE
APARTMENTS_AVG	TOTALAREA_MODE	0.8926	TOTALAREA_MODE
LIVINGAREA_AVG	APARTMENTS_MODE	0.8907	LIVINGAREA_AVG
APARTMENTS_MEDI	TOTALAREA_MODE	0.8866	TOTALAREA_MODE
LIVINGAPARTMENTS_MEDI	LIVINGAREA_MEDI	0.8847	LIVINGAREA_MEDI
LIVINGAREA_AVG	LIVINGAPARTMENTS_MEDI	0.8832	LIVINGAREA_AVG
LIVINGAPARTMENTS_AVG	LIVINGAREA_AVG	0.8808	LIVINGAREA_AVG
EXT_SOURCE_MEAN	EXT_SOURCE_MIN	0.8799	EXT_SOURCE_MEAN
CNT_CHILDREN	CNT_FAM_MEMBERS	0.8792	CNT_CHILDREN
LIVINGAPARTMENTS_MODE	LIVINGAREA_MODE	0.8785	LIVINGAREA_MODE
LIVINGAPARTMENTS_AVG	LIVINGAREA_MEDI	0.8783	LIVINGAREA_MEDI
LIVINGAPARTMENTS_MODE	LIVINGAREA_MEDI	0.8744	LIVINGAREA_MEDI
LIVINGAREA_AVG	LIVINGAPARTMENTS_MODE	0.8731	LIVINGAREA_AVG

col1	col2	abs_corr	prefer
ELEVATORS_MEDI	LIVINGAREA_MEDI	0.8683	ELEVATORS_MEDI
ELEVATORS_AVG	LIVINGAREA_AVG	0.8678	ELEVATORS_AVG
LIVINGAREA_AVG	ELEVATORS_MEDI	0.8658	ELEVATORS_MEDI
ELEVATORS_AVG	LIVINGAREA_MEDI	0.8656	ELEVATORS_AVG
APARTMENTS_MODE	TOTALAREA_MODE	0.8636	TOTALAREA_MODE
REG_REGION_NOT_WORK_REGION	LIVE_REGION_NOT_WORK_REGION	0.8606	REG_REGION_NOT_W ORK_REGION
DEF_30_CNT_SOCIAL_CIRCLE	DEF_60_CNT_SOCIAL_CIRCLE	0.8605	DEF_30_CNT_SOCIAL_ CIRCLE
LIVINGAREA_MODE	LIVINGAPARTMENTS_MEDI	0.8574	LIVINGAREA_MODE
ELEVATORS_MODE	LIVINGAREA_MODE	0.856	ELEVATORS_MODE
ELEVATORS_MODE	LIVINGAREA_MEDI	0.8558	LIVINGAREA_MEDI
LIVINGAREA_AVG	ELEVATORS_MODE	0.8526	LIVINGAREA_AVG
LIVINGAPARTMENTS_AVG	LIVINGAREA_MODE	0.8517	LIVINGAREA_MODE

6.7 Engineered Feature Correlation vs Components

feature	corr	outperforms_components
CREDIT_INCOME_RATIO	0.0077	False
ANNUITY_INCOME_RATIO	0.0143	True
CREDIT_GOODS_RATIO	0.0694	True
AGE_YEARS	0.0782	False
EMPLOYED_YEARS	0.075	False
EMPLOYMENT_RATIO	0.068	False
EXT_SOURCE_MEAN	0.2221	True
EXT_SOURCE_MIN	0.1853	True

6.8 Recommended Columns for Dropping (Phase 1)

column	reason
COMMONAREA_AVG	>60% null
COMMONAREA_MODE	>60% null
COMMONAREA_MEDI	>60% null
NONLIVINGAPARTMENTS_MEDI	>60% null
NONLIVINGAPARTMENTS_MODE	>60% null
NONLIVINGAPARTMENTS_AVG	>60% null
FONDKAPREMONT_MODE	>60% null
LIVINGAPARTMENTS_AVG	>60% null
LIVINGAPARTMENTS_MEDI	>60% null
LIVINGAPARTMENTS_MODE	>60% null
FLOORSMIN_MODE	>60% null
FLOORSMIN_AVG	>60% null
FLOORSMIN_MEDI	>60% null
YEARS_BUILD_AVG	>60% null
YEARS_BUILD_MODE	>60% null
YEARS_BUILD_MEDI	>60% null
OWN_CAR_AGE	>60% null

column	reason
AGE_YEARS	Redundant with DAYS_BIRTH (corr=1.0)
EMPLOYED_YEARS	Redundant with DAYS_EMPLOYED (corr=1.0)
OBS_60_CNT_SOCIAL_CIRCLE	Redundant with OBS_30_CNT_SOCIAL_CIRCLE (corr=0.9985)
FLOORSMAX_MEDI	Redundant with FLOORSMAX_AVG (corr=0.997)
ENTRANCES_MEDI	Redundant with ENTRANCES_AVG (corr=0.9969)
ELEVATORS_MEDI	Redundant with ELEVATORS_AVG (corr=0.9961)
LIVINGAREA_MEDI	Redundant with LIVINGAREA_AVG (corr=0.9956)
APARTMENTS_MEDI	Redundant with APARTMENTS_AVG (corr=0.9951)
BASEMENTAREA_MEDI	Redundant with BASEMENTAREA_AVG (corr=0.9943)
YEARS_BEGINEXPLUATATION_AVG	Redundant with YEARS_BEGINEXPLUATATION_MEDI (corr=0.9938)
LANDAREA_AVG	Redundant with LANDAREA_MEDI (corr=0.9916)
NONLIVINGAREA_MEDI	Redundant with NONLIVINGAREA_AVG (corr=0.9904)
FLOORSMAX_MODE	Redundant with FLOORSMAX_MEDI (corr=0.9882)
AMT_CREDIT	Redundant with AMT_GOODS_PRICE (corr=0.987)
ELEVATORS_MODE	Redundant with ELEVATORS_MEDI (corr=0.9828)
LANDAREA_MODE	Redundant with LANDAREA_MEDI (corr=0.9808)
ENTRANCES_MODE	Redundant with ENTRANCES_MEDI (corr=0.9807)
BASEMENTAREA_MODE	Redundant with BASEMENTAREA_MEDI (corr=0.9779)
APARTMENTS_MODE	Redundant with APARTMENTS_MEDI (corr=0.9772)
NONLIVINGAREA_MODE	Redundant with NONLIVINGAREA_MEDI (corr=0.9758)
LIVINGAREA_MODE	Redundant with LIVINGAREA_MEDI (corr=0.9747)
YEARS_BEGINEXPLUATATION_MODE	Redundant with YEARS_BEGINEXPLUATATION_AVG (corr=0.9719)
EMPLOYMENT_RATIO	Redundant with DAYS_EMPLOYED (corr=0.9547)
REGION_RATING_CLIENT	Redundant with REGION_RATING_CLIENT_W_CITY (corr=0.9508)
TOTALAREA_MODE	Redundant with LIVINGAREA_AVG (corr=0.925)
APARTMENTS_AVG	Redundant with LIVINGAREA_AVG (corr=0.9136)
EXT_SOURCE_MIN	Redundant with EXT_SOURCE_MEAN (corr=0.8799)
CNT_FAM_MEMBERS	Redundant with CNT_CHILDREN (corr=0.8792)
LIVINGAREA_AVG	Redundant with ELEVATORS_AVG (corr=0.8678)
LIVE_REGION_NOT_WORK_REGION	Redundant with REG_REGION_NOT_WORK_REGION (corr=0.8606)
DEF_60_CNT_SOCIAL_CIRCLE	Redundant with DEF_30_CNT_SOCIAL_CIRCLE (corr=0.8605)
SK_ID_CURR	ID column – non-predictive

Based on >60% nulls, high multicollinearity with a stronger predictor, or non-predictive metadata.

Section 7 — Pipeline Validation (Phase 3)

7.1 Row Count Check

check	application_train	joined_df	status
Row count	307511	307511	PASS

7.2 Column Inventory by Source Table

source	n_columns
application_train	122
bureau	17
bureau_balance	7
credit_card	17
cross_table	4
installments	12
missingness_flag	6
pos_cash	12
prev_application	18

Column count per source table group after all joins.

7.3 Missingness Consistency Check

For each table: fraction of applicants with `has_*=0` should equal null rate of representative column within 1 percentage point.

has_flag	flag=0 rate	null_rate	diff	status
has_bureau_record	0.1431	0.1431	0	PASS
has_bureau_balance	0.7001	0.7001	0	PASS
has_prev_application	0.0535	0.0535	0	PASS
has_installments	0.0516	0.0516	0	PASS
has_pos_cash	0.0588	0.0588	0	PASS
has_credit_card	0.7174	0.7174	0	PASS

PASS = missingness flag and null rate agree within 1pp. FAIL indicates a pipeline join error.

7.4 Range Validation

column	min	max	expected	status
bureau_active_ratio	0	1	$\geq 0, \leq 1$	PASS
bureau_utilization_ratio	-3.112e+12	2.25e+12	$\geq 0, \leq 1.05$	FAIL
bureau_bal_dpd_rate	0	1	$\geq 0, \leq 1$	PASS
inst_late_rate	0	1	$\geq 0, \leq 1$	PASS
inst_underpay_rate	0	1	$\geq 0, \leq 1$	PASS
pos_dpd_rate	0	1	$\geq 0, \leq 1$	PASS
cc_dpd_rate	0	0.9895	$\geq 0, \leq 1$	PASS

column	min	max	expected	status
cc_avg_utilization	-3.706e+07	3.81e+11	>=0, <=1.05	FAIL
systemic_delinquency_score	0	4	>=0, <=4	PASS
bureau_bal_worst_ever_status	0	5	>=0, <=5	PASS
prev_approval_rate	0	1	>=0, <=1	PASS
pos_max_dpd	0	4231	>=0	PASS
cc_max_dpd	0	2831	>=0	PASS
inst_max_days_late	-156	2884	>=-365, <=365	FAIL

All columns must fall within expected ranges. FAIL indicates an aggregation error.

Section 8 — Supplementary Feature Signal (Phase 3)

8.1 Top 50 Features by |Correlation with TARGET| — Full Joined Dataset

rank	feature	abs_corr	source
1	TARGET	1	application_train
2	EXT_SOURCE_3	0.1789	application_train
3	EXT_SOURCE_2	0.1605	application_train
4	EXT_SOURCE_1	0.1553	application_train
5	bureau_recent_inquiry_count	0.0901	bureau
6	bureau_avg_credit_age	0.0897	bureau
7	cc_avg_balance	0.0872	credit_card
8	cc_any_atm_drawings	0.0785	credit_card
9	DAYS_BIRTH	0.0782	application_train
10	bureau_active_ratio	0.0774	bureau
11	DAYS_EMPLOYED	0.075	application_train
12	inst_late_rate	0.07	installments
13	cc_max_balance	0.0688	credit_card
14	bureau_active_count	0.0671	bureau
15	prev_refused_count	0.0645	prev_application
16	prev_approval_rate	0.0635	prev_application
17	inst_underpay_rate	0.0626	installments
18	cc_above_minimum_rate	0.0623	credit_card
19	REGION_RATING_CLIENT_W_CITY	0.0609	application_train
20	cc_months_count	0.0605	credit_card
21	internal_dpd_composite	0.0602	cross_table
22	cc_avg_atm_drawings	0.0599	credit_card
23	REGION_RATING_CLIENT	0.0589	application_train
24	bureau_bal_dpd_rate	0.0589	bureau_balance
25	DAYS_LAST_PHONE_CHANGE	0.0552	application_train
26	bureau_has_microfinance	0.0536	bureau
27	DAYS_ID_PUBLISH	0.0515	application_train
28	REG_CITY_NOT_WORK_CITY	0.051	application_train
29	inst_any_late	0.0494	installments
30	systemic_delinquency_score	0.0488	cross_table
31	prev_avg_decision_days	0.0469	prev_application
32	FLAG_EMP_PHONE	0.046	application_train
33	REG_CITY_NOT_LIVE_CITY	0.0444	application_train
34	FLAG_DOCUMENT_3	0.0443	application_train
35	bureau_bal_total_months	0.044	bureau_balance
36	FLOORSMAX_AVG	0.044	application_train
37	FLOORSMAX_MEDI	0.0438	application_train
38	FLOORSMAX_MODE	0.0432	application_train
39	DAYS_REGISTRATION	0.042	application_train

rank	feature	abs_corr	source
40	AMT_GOODS_PRICE	0.0396	application_train
41	cc_total_atm_drawings	0.0381	credit_card
42	OWN_CAR_AGE	0.0376	application_train
43	REGION_POPULATION_RELATIVE	0.0372	application_train
44	bureau_bal_worst_ever_status	0.0364	bureau_balance
45	pos_active_count	0.0359	pos_cash
46	pos_months_count	0.0356	pos_cash
47	prev_avg_annuity	0.0349	prev_application
48	ELEVATORS_AVG	0.0342	application_train
49	ELEVATORS_MEDI	0.0339	application_train
50	pos_any_dpd	0.0339	pos_cash

Top 50 features by absolute Pearson correlation with TARGET across all columns in the joined dataset. Source table annotation shows which data source each feature originates from.

8.2 Top 10 Supplementary Features (non-application_train)

rank	feature	abs_corr	source
5	bureau_recent_inquiry_count	0.0901	bureau
6	bureau_avg_credit_age	0.0897	bureau
7	cc_avg_balance	0.0872	credit_card
8	cc_any_atm_drawings	0.0785	credit_card
10	bureau_active_ratio	0.0774	bureau
12	inst_late_rate	0.07	installments
13	cc_max_balance	0.0688	credit_card
14	bureau_active_count	0.0671	bureau
15	prev_refused_count	0.0645	prev_application
16	prev_approval_rate	0.0635	prev_application

Top 10 features from supplementary tables only — demonstrates the predictive value added by feature engineering over the base application data.

8.3 Key Feature Rank Positions

feature	rank	abs_corr
inst_late_rate	12	0.07
bureau_bal_worst_ever_status	44	0.0364
systemic_delinquency_score	30	0.0488
internal_vs_external_dpd_diff	90	0.0211

Rank positions of the highest-signal cross-table features in the full feature ranking. Low rank numbers confirm the aggregation strategy produced valuable features.

8.4 Cross-Table Feature Validation

cross_feature	corr	recommendation
systemic_delinquency_score	0.0488	KEEP
internal_vs_external_dpd_diff	0.0211	DROP candidate

cross_feature	corr	recommendation
pos_dpd_trend	0.006	KEEP
cc_dpd_trend	0.0034	KEEP
cc_utilization_trend	0.0034	KEEP
credit_request_ratio	0.0047	DROP candidate
annuity_request_ratio	0.0006	DROP candidate
inst_std_days_late	0.0043	DROP candidate

Each cross-table feature compared to its component features. KEEP = feature adds value over components. DROP candidate = components are stronger.

Section 9 — Missingness Structure (Phase 3)

9.1 Default Rate by has_* Flag Value

flag	default_rate_0	default_rate_1	abs_diff	informative
has_bureau_record	0.1012	0.0773	0.0239	YES
has_prev_application	0.0596	0.0819	0.0224	YES
has_installments	0.0598	0.0819	0.0221	YES
has_pos_cash	0.0666	0.0816	0.015	no
has_credit_card	0.0784	0.0867	0.0083	no
has_bureau_balance	0.0804	0.0814	0.001	no

Default rate when applicant has no record (flag=0) vs has a record (flag=1). "informative=YES" means absence/presence predicts default by >2pp — retain as a feature.

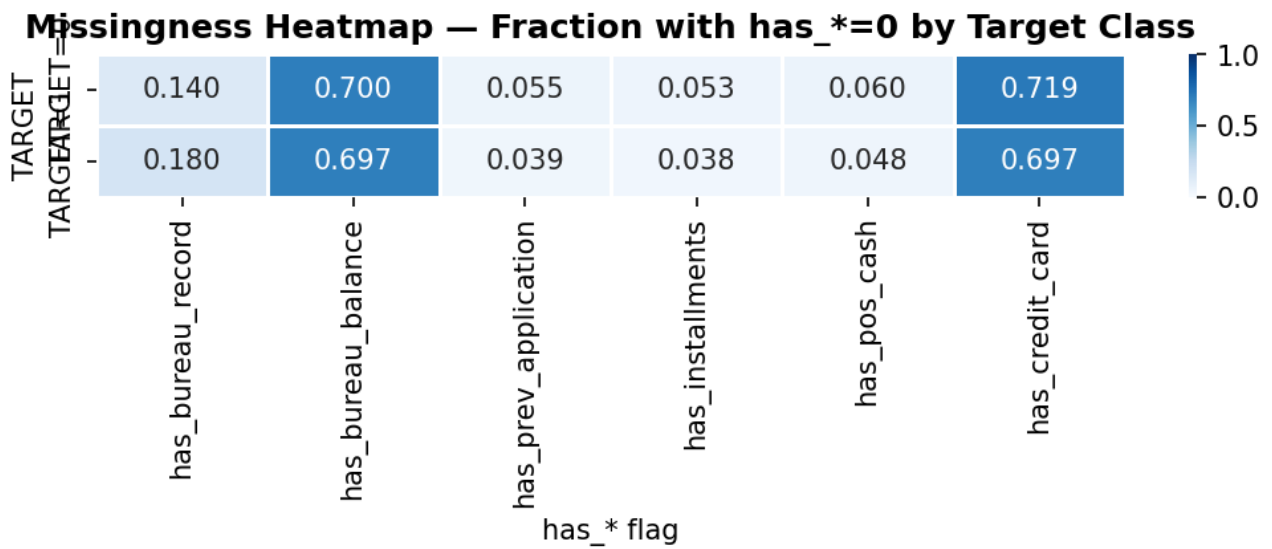
9.2 no_history_count Distribution and Default Rate

Correlation of no_history_count with TARGET: **-0.0067**

no_history_count	n_applicants	default_rate
0	2.893e+04	0.0848
1	1.033e+05	0.0821
2	1.284e+05	0.0758
3	3.143e+04	0.1034
4	3742	0.0612
5	9565	0.0553
6	2200	0.0714

Distribution of no_history_count (number of supplementary tables with no record). no_history_count=max means no history in any supplementary table — first-time borrowers.

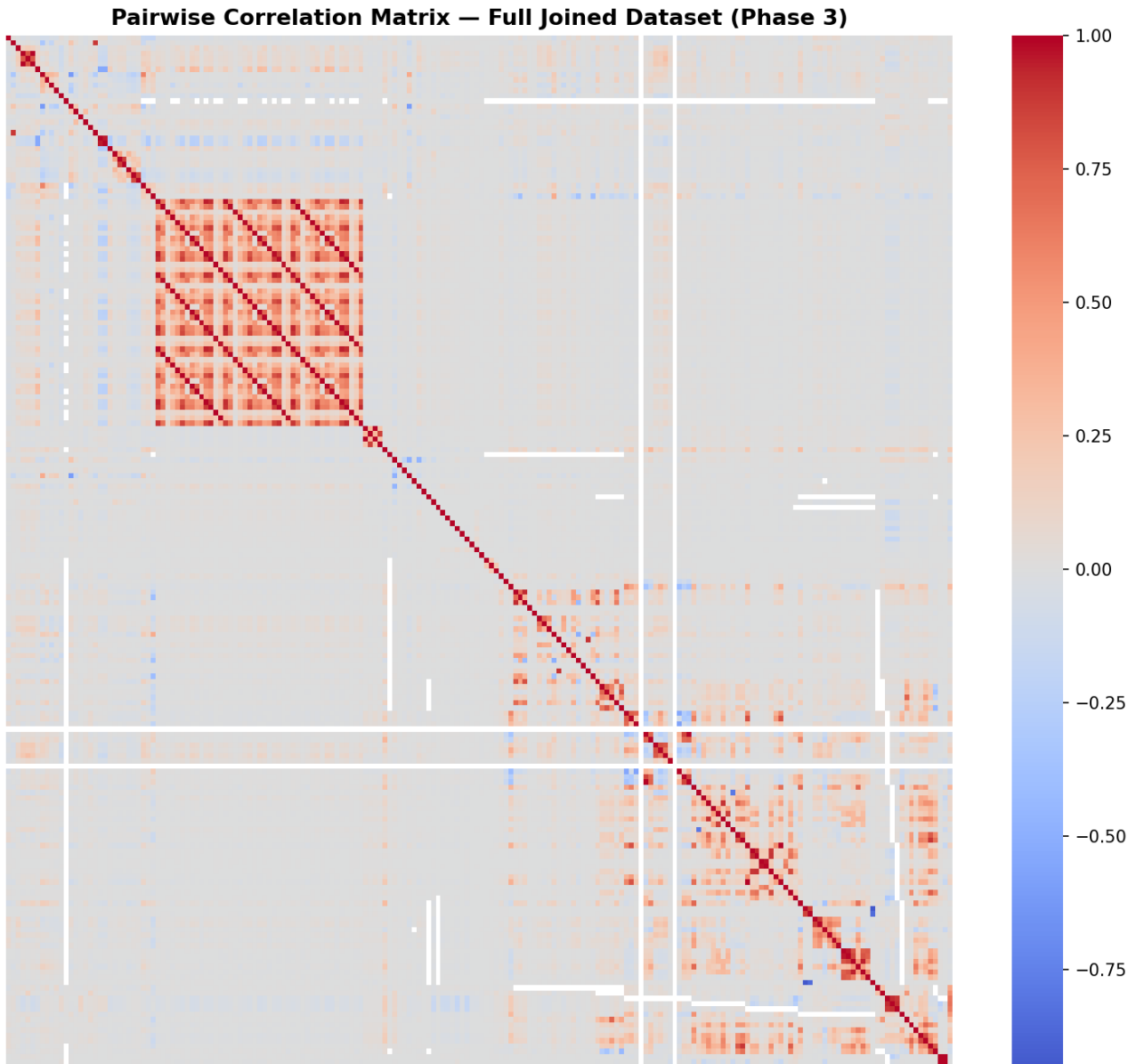
9.3 Missingness Heatmap



Fraction of applicants with has_=0 (no record in that table) split by target class. Differences between TARGET=0 and TARGET=1 rows confirm missingness is informative.*

Section 10 — Correlation Structure (Phase 3)

10.1 Full Pairwise Correlation Heatmap — Joined Dataset



10.2 High-Correlation Pairs ($|\text{corr}| > 0.85$) — Joined Dataset

col1	source1	col2	source2	abs_corr	prefer
prev_most_recent_decision	prev_application	days_since_last_application	application_train	1	prev_most_recent_decision

col1	source1	col2	source2	abs_corr	prefer
credit_request_ratio	application_train	annuity_request_ratio	application_train	1	credit_request_ratio
YEARS_BUILD_AVG	application_train	YEARS_BUILD_MEDI	application_train	0.9985	YEARS_BUILD_MEDI
OBS_30_CNT_SOCIAL_CIRCLE	application_train	OBS_60_CNT_SOCIAL_CIRCLE	application_train	0.9985	OBS_30_CNT_SOCIAL_CIRCLE
FLOORS_MIN_AVG	application_train	FLOORS_MIN_MEDI	application_train	0.9972	FLOORS_MIN_AVG
FLOORS_MAX_AVG	application_train	FLOORS_MAX_MEDI	application_train	0.997	FLOORS_MAX_AVG
ENTRANCES_AVG	application_train	ENTRANCES_MEDI	application_train	0.9969	ENTRANCES_AVG
ELEVATORS_AVG	application_train	ELEVATORS_MEDI	application_train	0.9961	ELEVATORS_AVG
COMMONAREA_AVG	application_train	COMMONAREA_MEDI	application_train	0.996	COMMONAREA_MEDI
LIVINGAREA_AVG	application_train	LIVINGAREA_MEDI	application_train	0.9956	LIVINGAREA_AVG
APARTMENTS_AVG	application_train	APARTMENTS_MEDI	application_train	0.9951	APARTMENTS_AVG
BASEMENTAREA_AVG	application_train	BASEMENTAREA_MEDI	application_train	0.9943	BASEMENTAREA_AVG
YEARS_BEGINEXPLUATION_AVG	application_train	YEARS_BEGINEXPLUATION_MEDI	application_train	0.9938	YEARS_BEGINEXPLUATION_MEDI
LIVINGAPARTMENTS_AVG	application_train	LIVINGAPARTMENTS_MEDI	application_train	0.9938	LIVINGAPARTMENTS_AVG
pos_months_count	pos_cash	pos_active_count	pos_cash	0.9917	pos_active_count
LANDAREA_AVG	application_train	LANDAREA_MEDI	application_train	0.9916	LANDAREA_MEDI
NONLIVINGAPARTMENTS_AVG	application_train	NONLIVINGAPARTMENTS_MEDI	application_train	0.9908	NONLIVINGAPARTMENTS_AVG
NONLIVINGAREA_AVG	application_train	NONLIVINGAREA_MEDI	application_train	0.9904	NONLIVINGAREA_AVG
YEARS_BUILD_MODE	application_train	YEARS_BUILD_MEDI	application_train	0.9895	YEARS_BUILD_MEDI
YEARS_BUILD_AVG	application_train	YEARS_BUILD_MODE	application_train	0.9894	YEARS_BUILD_AVG

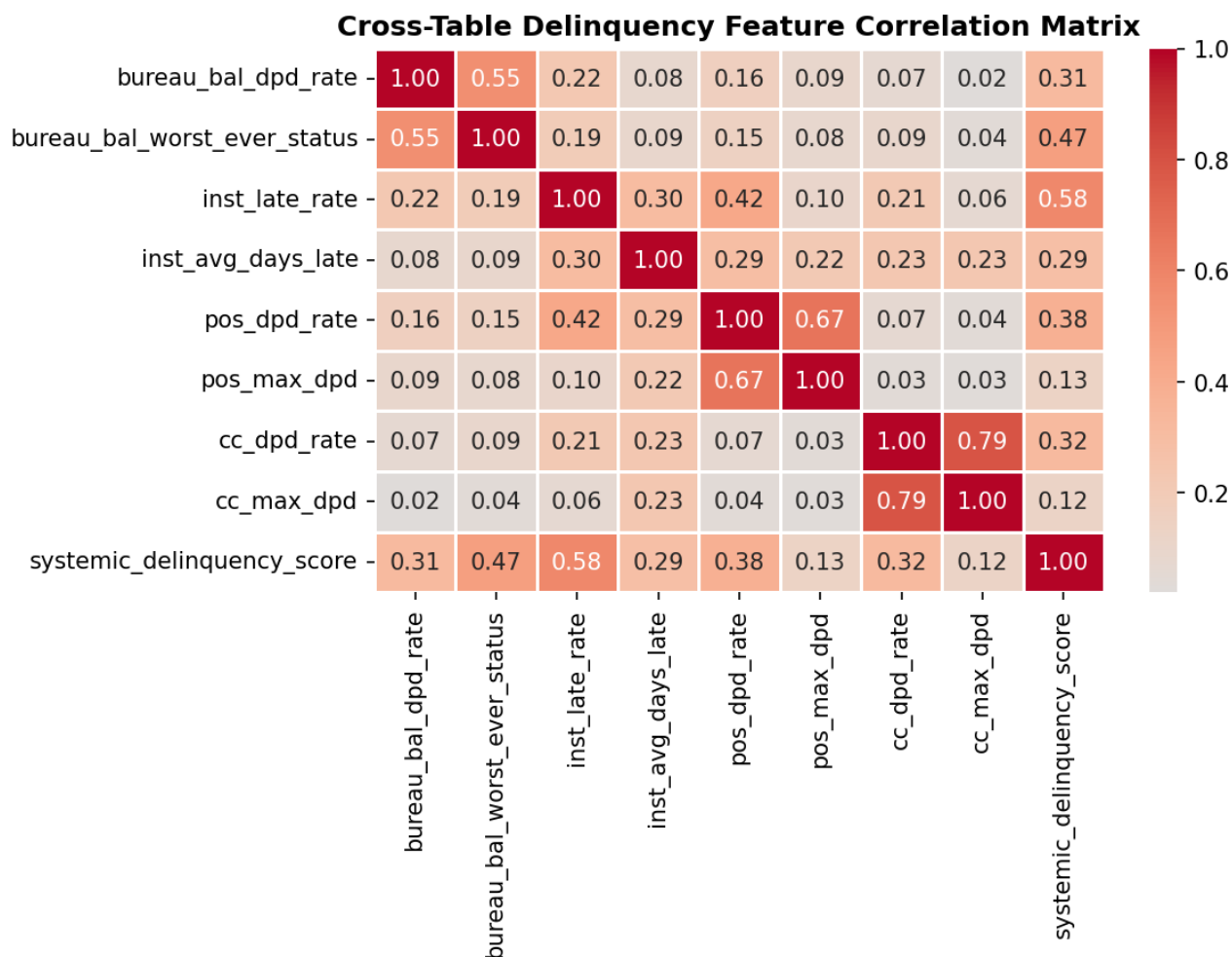
col1	source1	col2	source2	abs_corr	prefer
FLOORSMIN_MODE	application_train	FLOORSMIN_MEDI	application_train	0.9884	FLOORS MIN_MEDI
FLOORSMAX_MODE	application_train	FLOORSMAX_MEDI	application_train	0.9882	FLOORS MAX_MEDI
AMT_CREDIT	application_train	AMT_GOODS_PRICE	application_train	0.987	AMT_GOODS_PRICE
FLOORSMIN_AVG	application_train	FLOORSMIN_MODE	application_train	0.9859	FLOORS MIN_AVG
FLOORSMAX_AVG	application_train	FLOORSMAX_MODE	application_train	0.9857	FLOORS MAX_AVG
ELEVATORS_MODE	application_train	ELEVATORS_MEDI	application_train	0.9828	ELEVATORS_MEDI
LANDAREA_MODE	application_train	LANDAREA_MEDI	application_train	0.9808	LANDAREA_MEDI
ENTRANCES_MODE	application_train	ENTRANCES_MEDI	application_train	0.9807	ENTRANCES_MEDI
COMMONAREA_MODE	application_train	COMMONAREA_MEDI	application_train	0.9799	COMMON AREA_MEDI
ELEVATORS_AVG	application_train	ELEVATORS_MODE	application_train	0.9788	ELEVATORS_AVG
NONLIVINGAPARTMENTS_MODE	application_train	NONLIVINGAPARTMENTS_MEDI	application_train	0.9786	NONLIVINGAPARTMENTS_MEDI
BASEMENTAREA_MODE	application_train	BASEMENTAREA_MEDI	application_train	0.9779	BASEMENTAREA_MEDI
ENTRANCES_AVG	application_train	ENTRANCES_MODE	application_train	0.9777	ENTRANCES_AVG
APARTMENTS_MODE	application_train	APARTMENTS_MEDI	application_train	0.9772	APARTMENTS_MEDI
COMMONAREA_AVG	application_train	COMMONAREA_MODE	application_train	0.9771	COMMON AREA_AVG
NONLIVINGAREA_MODE	application_train	NONLIVINGAREA_MEDI	application_train	0.9758	NONLIVINGAREA_MEDI
LIVINGAPARTMENTS_MODE	application_train	LIVINGAPARTMENTS_MEDI	application_train	0.9756	LIVINGAPARTMENTS_MEDI
LIVINGAREA_MODE	application_train	LIVINGAREA_MEDI	application_train	0.9747	LIVINGAREA_MEDI
LANDAREA_AVG	application_train	LANDAREA_MODE	application_train	0.9737	LANDAREA_AVG
BASEMENTAREA_AVG	application_train	BASEMENTAREA_MODE	application_train	0.9735	BASEMENTAREA_AVG

col1	source1	col2	source2	abs_corr	prefer
APARTMENTS_AVG	application_train	APARTMENTS_MODE	application_train	0.9733	APARTMENTS_AVG
LIVINGAREA_AVG	application_train	LIVINGAREA_MODE	application_train	0.972	LIVINGAREA_AVG
YEARS_BEGINEXPLUATION_AVG	application_train	YEARS_BEGINEXPLUATION_MODE	application_train	0.9719	YEARS_BEGINEXPLUATION_AVG
LIVINGAPARTMENTS_AVG	application_train	LIVINGAPARTMENTS_MODE	application_train	0.9701	LIVINGAPARTMENTS_AVG
NONLIVINGAPARTMENTS_AVG	application_train	NONLIVINGAPARTMENTS_MODE	application_train	0.9694	NONLIVINGAPARTMENTS_AVG
NONLIVINGAREA_AVG	application_train	NONLIVINGAREA_MODE	application_train	0.9661	NONLIVINGAREA_AVG
pos_avg_dpd_def	pos_cash	pos_max_dpd_def	pos_cash	0.9649	pos_max_dpd_def
YEARS_BEGINEXPLUATION_MODE	application_train	YEARS_BEGINEXPLUATION_MEDI	application_train	0.9635	YEARS_BEGINEXPLUATION_MEDI
cc_avg_dpd	credit_card	cc_max_dpd	credit_card	0.9605	cc_max_dpd
REGION_RATING_CLIENT	application_train	REGION_RATING_CLIENT_W_CITY	application_train	0.9508	REGION_RATING_CLIENT_W_CITY
APARTMENTS_AVG	application_train	LIVINGAPARTMENTS_AVG	application_train	0.944	APARTMENTS_AVG
APARTMENTS_MEDI	application_train	LIVINGAPARTMENTS_MEDI	application_train	0.9425	APARTMENTS_MEDI
cc_dpd_month_count	credit_card	cc_dpd_rate	credit_card	0.9422	cc_dpd_month_count
APARTMENTS_AVG	application_train	LIVINGAPARTMENTS_MEDI	application_train	0.9419	APARTMENTS_AVG
cc_avg_utilization	credit_card	cc_utilization_trend	credit_card	0.938	cc_utilization_trend
APARTMENTS_MODE	application_train	LIVINGAPARTMENTS_MODE	application_train	0.9378	APARTMENTS_MODE
LIVINGAPARTMENTS_AVG	application_train	APARTMENTS_MEDI	application_train	0.9356	APARTMENTS_MEDI
LIVINGAPARTMENTS_MODE	application_train	APARTMENTS_MEDI	application_train	0.9322	APARTMENTS_MEDI

col1	source1	col2	source2	abs_corr	prefer
APARTMENTS_AVG	application_train	LIVINGAPARTMENTS_MODE	application_train	0.9306	APARTMENTS_AVG
LIVINGAREA_AVG	application_train	TOTALAREA_MODE	application_train	0.925	LIVINGAREA_AVG

Column pairs with $|\text{correlation}| > 0.85$ in the joined dataset. "prefer" indicates the column with higher absolute correlation with TARGET — the one to retain if one must be dropped. Showing up to 60 pairs.

10.3 Cross-Table Delinquency Feature Correlation Matrix



Pairwise correlation among the 9 key delinquency features from different source tables. High correlation (>0.85) means two features capture the same underlying signal — the weaker predictor is a drop candidate.

feature	bureau_bal_dpd_rate	bureau_bal_worst_ever_status	inst_late_rate	inst_avg_days_late	pos_dpd_rate	pos_max_dpd	cc_dpd_rate	cc_max_dpd	systemic_delinquency_score
bureau_bal_dpd_rate	1	0.5549	0.2209	0.0841	0.16	0.0875	0.066	0.0207	0.314
bureau_bal_worst_ever_status	0.5549	1	0.1925	0.0932	0.1462	0.0787	0.0868	0.0363	0.4738
inst_late_rate	0.2209	0.1925	1	0.3038	0.4243	0.0994	0.2131	0.0631	0.582

feature	bureau_bal_dpd_rate	bureau_bal_worst_status	inst_late_rate	inst_avg_days_late	pos_dpd_rate	pos_max_dpd	cc_dpd_rate	cc_max_dpd	systemic_delinquency_score
inst_avg_days_late	0.0841	0.0932	0.3038	1	0.2938	0.2196	0.232	0.2339	0.2879
pos_dpd_rate	0.16	0.1462	0.4243	0.2938	1	0.6684	0.065	0.0373	0.3805
pos_max_dpd	0.0875	0.0787	0.0994	0.2196	0.6684	1	0.0332	0.0282	0.1285
cc_dpd_rate	0.066	0.0868	0.2131	0.232	0.065	0.0332	1	0.7945	0.3207
cc_max_dpd	0.0207	0.0363	0.0631	0.2339	0.0373	0.0282	0.7945	1	0.1243
systemic_delinquency_score	0.314	0.4738	0.582	0.2879	0.3805	0.1285	0.3207	0.1243	1

Numeric cross-table delinquency correlation matrix.

10.4 Within-Table Redundancy (pairs with |corr| > 0.90)

table	col1	col2	abs_corr
pos_cash	pos_months_count	pos_active_count	0.9917
pos_cash	pos_avg_dpd_def	pos_max_dpd_def	0.9649
credit_card	cc_avg_dpd	cc_max_dpd	0.9605
credit_card	cc_dpd_month_count	cc_dpd_rate	0.9422
credit_card	cc_avg_utilization	cc_utilization_trend	0.938
pos_cash	pos_avg_dpd	pos_max_dpd	0.9244
bureau	bureau_loan_count	bureau_closed_count	0.9237

Within-table column pairs with absolute correlation > 0.90. These are near-redundant and one column from each pair can be dropped.

Section 11 — Phase 3 Output Summary

11.1 Row Count Confirmation

check	application_train	joined_df	status
Row count	307511	307511	PASS

11.2 Column Count by Source Table

source	n_columns
application_train	122
bureau	17
bureau_balance	7
credit_card	17
cross_table	4
installments	12
missingness_flag	6
pos_cash	12
prev_application	18

11.3 Missingness Consistency Check Results

has_flag	flag=0 rate	null_rate	diff	status
has_bureau_record	0.1431	0.1431	0	PASS
has_bureau_balance	0.7001	0.7001	0	PASS
has_prev_application	0.0535	0.0535	0	PASS
has_installments	0.0516	0.0516	0	PASS
has_pos_cash	0.0588	0.0588	0	PASS
has_credit_card	0.7174	0.7174	0	PASS

11.4 Range Validation Results

column	min	max	expected	status
bureau_active_ratio	0	1	>=0, <=1	PASS
bureau_utilization_ratio	-3.112e+12	2.25e+12	>=0, <=1.05	FAIL
bureau_bal_dpd_rate	0	1	>=0, <=1	PASS
inst_late_rate	0	1	>=0, <=1	PASS
inst_underpay_rate	0	1	>=0, <=1	PASS
pos_dpd_rate	0	1	>=0, <=1	PASS
cc_dpd_rate	0	0.9895	>=0, <=1	PASS
cc_avg_utilization	-3.706e+07	3.81e+11	>=0, <=1.05	FAIL
systemic_delinquency_score	0	4	>=0, <=4	PASS
bureau_bal_worst_ever_status	0	5	>=0, <=5	PASS
prev_approval_rate	0	1	>=0, <=1	PASS
pos_max_dpd	0	4231	>=0	PASS

column	min	max	expected	status
cc_max_dpd	0	2831	≥ 0	PASS
inst_max_days_late	-156	2884	$\geq -365, \leq 365$	FAIL

11.5 Top 10 Supplementary Features

rank	feature	abs_corr	source
5	bureau_recent_inquiry_count	0.0901	bureau
6	bureau_avg_credit_age	0.0897	bureau
7	cc_avg_balance	0.0872	credit_card
8	cc_any_atm_drawings	0.0785	credit_card
10	bureau_active_ratio	0.0774	bureau
12	inst_late_rate	0.07	installments
13	cc_max_balance	0.0688	credit_card
14	bureau_active_count	0.0671	bureau
15	prev_refused_count	0.0645	prev_application
16	prev_approval_rate	0.0635	prev_application

11.6 Cross-Table Feature Validation (Keep / Drop)

cross_feature	corr	recommendation
systemic_delinquency_score	0.0488	KEEP
internal_vs_external_dpd_diff	0.0211	DROP candidate
pos_dpd_trend	0.006	KEEP
cc_dpd_trend	0.0034	KEEP
cc_utilization_trend	0.0034	KEEP
credit_request_ratio	0.0047	DROP candidate
annuity_request_ratio	0.0006	DROP candidate
inst_std_days_late	0.0043	DROP candidate

11.7 Informative has_* Flags ($|\Delta \text{ default rate}| > 0.02$)

flag	default_rate_0	default_rate_1	abs_diff
has_bureau_record	0.1012	0.0773	0.0239
has_prev_application	0.0596	0.0819	0.0224
has_installments	0.0598	0.0819	0.0221

11.8 no_history_count Distribution and Default Rate

no_history_count	n_applicants	default_rate
0	2.893e+04	0.0848
1	1.033e+05	0.0821
2	1.284e+05	0.0758
3	3.143e+04	0.1034
4	3742	0.0612
5	9565	0.0553
6	2200	0.0714

11.9 All Column Pairs with |Corr| > 0.85 (Joined Dataset, top 40)

col1	source1	col2	source2	abs_corr	prefer
prev_most_recent_decision	prev_application	days_since_last_application	application_train	1	prev_most_recent_decision
credit_request_ratio	application_train	annuity_request_ratio	application_train	1	credit_request_ratio
YEARS_BUILD_AVG	application_train	YEARS_BUILD_MEDI	application_train	0.9985	YEARS_BUILD_MEDI
OBS_30_CNT_SOCIAL_CIRCLE	application_train	OBS_60_CNT_SOCIAL_CIRCLE	application_train	0.9985	OBS_30_CNT_SOCIAL_CIRCLE
FLOORSMIN_AVG	application_train	FLOORSMIN_MEDI	application_train	0.9972	FLOORSMIN_AVG
FLOORSMAX_AVG	application_train	FLOORSMAX_MEDI	application_train	0.997	FLOORSMAX_AVG
ENTRANCES_AVG	application_train	ENTRANCES_MEDI	application_train	0.9969	ENTRANCES_AVG
ELEVATORS_AVG	application_train	ELEVATORS_MEDI	application_train	0.9961	ELEVATORS_AVG
COMMONAREA_AVG	application_train	COMMONAREA_MEDI	application_train	0.996	COMMONAREA_MEDI
LIVINGAREA_AVG	application_train	LIVINGAREA_MEDI	application_train	0.9956	LIVINGAREA_AVG
APARTMENTS_AVG	application_train	APARTMENTS_MEDI	application_train	0.9951	APARTMENTS_AVG
BASEMENTAREA_AVG	application_train	BASEMENTAREA_MEDI	application_train	0.9943	BASEMENTAREA_AVG
YEARS_BEGINEXPLUATION_AVG	application_train	YEARS_BEGINEXPLUATION_MEDI	application_train	0.9938	YEARS_BEGINEXPLUATION_MEDI
LIVINGAPARTMENTS_AVG	application_train	LIVINGAPARTMENTS_MEDI	application_train	0.9938	LIVINGAPARTMENTS_AVG
pos_months_count	pos_cash	pos_active_count	pos_cash	0.9917	pos_active_count
LANDAREA_AVG	application_train	LANDAREA_MEDI	application_train	0.9916	LANDAREA_MEDI
NONLIVINGAPARTMENTS_AVG	application_train	NONLIVINGAPARTMENTS_MEDI	application_train	0.9908	NONLIVINGAPARTMENTS_AVG
NONLIVINGAREA_AVG	application_train	NONLIVINGAREA_MEDI	application_train	0.9904	NONLIVINGAREA_AVG

col1	source1	col2	source2	abs_corr	prefer
YEARS_BUILD_MODE	application_train	YEARS_BUILD_MEDI	application_train	0.9895	YEARS_BUILD_MEDI
YEARS_BUILD_AVG	application_train	YEARS_BUILD_MODE	application_train	0.9894	YEARS_BUILD_AVG
FLOORSMIN_MODE	application_train	FLOORSMIN_MEDI	application_train	0.9884	FLOORSMIN_MEDI
FLOORSMAX_MODE	application_train	FLOORSMAX_MEDI	application_train	0.9882	FLOORSMAX_MEDI
AMT_CREDIT	application_train	AMT_GOODS_PRICE	application_train	0.987	AMT_GOODS_PRICE
FLOORSMIN_AVG	application_train	FLOORSMIN_MODE	application_train	0.9859	FLOORSMIN_AVG
FLOORSMAX_AVG	application_train	FLOORSMAX_MODE	application_train	0.9857	FLOORSMAX_AVG
ELEVATORS_MODE	application_train	ELEVATORS_MEDI	application_train	0.9828	ELEVATORS_MEDI
LANDAREA_MODE	application_train	LANDAREA_MEDI	application_train	0.9808	LANDAREA_MEDI
ENTRANCES_MODE	application_train	ENTRANCES_MEDI	application_train	0.9807	ENTRANCES_MEDI
COMMONAREA_MODE	application_train	COMMONAREA_MEDI	application_train	0.9799	COMMONAREA_MEDI
ELEVATORS_AVG	application_train	ELEVATORS_MODE	application_train	0.9788	ELEVATORS_AVG
NONLIVINGAPARTMENTS_MODE	application_train	NONLIVINGAPARTMENTS_MEDI	application_train	0.9786	NONLIVINGAPARTMENTS_MEDI
BASEMENTAREA_MODE	application_train	BASEMENTAREA_MEDI	application_train	0.9779	BASEMENTAREA_MEDI
ENTRANCES_AVG	application_train	ENTRANCES_MODE	application_train	0.9777	ENTRANCES_AVG
APARTMENTS_MODE	application_train	APARTMENTS_MEDI	application_train	0.9772	APARTMENTS_MEDI
COMMONAREA_AVG	application_train	COMMONAREA_MODE	application_train	0.9771	COMMONAREA_AVG
NONLIVINGAREA_MODE	application_train	NONLIVINGAREA_MEDI	application_train	0.9758	NONLIVINGAREA_MEDI
LIVINGAPARTMENTS_MODE	application_train	LIVINGAPARTMENTS_MEDI	application_train	0.9756	LIVINGAPARTMENTS_MEDI

col1	source1	col2	source2	abs_corr	prefer
LIVINGAREA_MODE	application_train	LIVINGAREA_MEDI	application_train	0.9747	LIVINGAREA_MEDI
LANDAREA_AVG	application_train	LANDAREA_MODE	application_train	0.9737	LANDAREA_AVG
BASEMENTAREA_AVG	application_train	BASEMENTAREA_MODE	application_train	0.9735	BASEMENTAREA_AVG

11.10 Final Drop List — Columns Recommended for Removal

column	reason
internal_vs_external_dpd_diff	Cross-table feature does not outperform components
credit_request_ratio	Cross-table feature does not outperform components
annuity_request_ratio	Cross-table feature does not outperform components
inst_std_days_late	Cross-table feature does not outperform components
days_since_last_application	Redundant with prev_most_recent_decision (corr=1.0)
YEARS_BUILD_AVG	Redundant with YEARS_BUILD_MEDI (corr=0.9985)
OBS_60_CNT_SOCIAL_CIRCLE	Redundant with OBS_30_CNT_SOCIAL_CIRCLE (corr=0.9985)
FLOORSMIN_MEDI	Redundant with FLOORSMIN_AVG (corr=0.9972)
FLOORSMAX_MEDI	Redundant with FLOORSMAX_AVG (corr=0.997)
ENTRANCES_MEDI	Redundant with ENTRANCES_AVG (corr=0.9969)
ELEVATORS_MEDI	Redundant with ELEVATORS_AVG (corr=0.9961)
COMMONAREA_AVG	Redundant with COMMONAREA_MEDI (corr=0.996)
LIVINGAREA_MEDI	Redundant with LIVINGAREA_AVG (corr=0.9956)
APARTMENTS_MEDI	Redundant with APARTMENTS_AVG (corr=0.9951)
BASEMENTAREA_MEDI	Redundant with BASEMENTAREA_AVG (corr=0.9943)
YEARS_BEGINEXPLUATATION_AVG	Redundant with YEARS_BEGINEXPLUATATION_MEDI (corr=0.9938)
LIVINGAPARTMENTS_MEDI	Redundant with LIVINGAPARTMENTS_AVG (corr=0.9938)
pos_months_count	Redundant with pos_active_count (corr=0.9917)
LANDAREA_AVG	Redundant with LANDAREA_MEDI (corr=0.9916)
NONLIVINGAPARTMENTS_MEDI	Redundant with NONLIVINGAPARTMENTS_AVG (corr=0.9908)
NONLIVINGAREA_MEDI	Redundant with NONLIVINGAREA_AVG (corr=0.9904)
YEARS_BUILD_MODE	Redundant with YEARS_BUILD_MEDI (corr=0.9895)
FLOORSMIN_MODE	Redundant with FLOORSMIN_MEDI (corr=0.9884)
FLOORSMAX_MODE	Redundant with FLOORSMAX_MEDI (corr=0.9882)
AMT_CREDIT	Redundant with AMT_GOODS_PRICE (corr=0.987)
ELEVATORS_MODE	Redundant with ELEVATORS_MEDI (corr=0.9828)
LANDAREA_MODE	Redundant with LANDAREA_MEDI (corr=0.9808)
ENTRANCES_MODE	Redundant with ENTRANCES_MEDI (corr=0.9807)
COMMONAREA_MODE	Redundant with COMMONAREA_MEDI (corr=0.9799)
NONLIVINGAPARTMENTS_MODE	Redundant with NONLIVINGAPARTMENTS_MEDI (corr=0.9786)
BASEMENTAREA_MODE	Redundant with BASEMENTAREA_MEDI (corr=0.9779)
APARTMENTS_MODE	Redundant with APARTMENTS_MEDI (corr=0.9772)
NONLIVINGAREA_MODE	Redundant with NONLIVINGAREA_MEDI (corr=0.9758)
LIVINGAPARTMENTS_MODE	Redundant with LIVINGAPARTMENTS_MEDI (corr=0.9756)
LIVINGAREA_MODE	Redundant with LIVINGAREA_MEDI (corr=0.9747)

column	reason
YEARS_BEGINEXPLUATATION_MODE	Redundant with YEARS_BEGINEXPLUATATION_AVG (corr=0.9719)
pos_avg_dpd_def	Redundant with pos_max_dpd_def (corr=0.9649)
cc_avg_dpd	Redundant with cc_max_dpd (corr=0.9605)
REGION_RATING_CLIENT	Redundant with REGION_RATING_CLIENT_W_CITY (corr=0.9508)
LIVINGAPARTMENTS_AVG	Redundant with APARTMENTS_AVG (corr=0.944)
cc_dpd_rate	Redundant with cc_dpd_month_count (corr=0.9422)
cc_avg_utilization	Redundant with cc_utilization_trend (corr=0.938)
TOTALAREA_MODE	Redundant with LIVINGAREA_AVG (corr=0.925)
pos_max_dpd	Redundant with pos_avg_dpd (corr=0.9244)
bureau_loan_count	Redundant with bureau_closed_count (corr=0.9237)
has_installments	Redundant with has_prev_application (corr=0.9206)
APARTMENTS_AVG	Redundant with LIVINGAREA_AVG (corr=0.9136)
has_pos_cash	Redundant with has_installments (corr=0.9066)
inst_underpay_count	Redundant with inst_late_count (corr=0.8992)
bureau_total_prolonged	Redundant with bureau_any_prolonged (corr=0.8965)
internal_dpd_composite	Redundant with inst_late_rate (corr=0.8925)
pos_avg_dpd	Redundant with pos_dpd_month_count (corr=0.8906)
CNT_FAM_MEMBERS	Redundant with CNT_CHILDREN (corr=0.8792)
cc_max_balance	Redundant with cc_avg_balance (corr=0.8752)
LIVINGAREA_AVG	Redundant with ELEVATORS_AVG (corr=0.8678)
LIVE_REGION_NOT_WORK_REGION	Redundant with REG_REGION_NOT_WORK_REGION (corr=0.8606)
DEF_60_CNT_SOCIAL_CIRCLE	Redundant with DEF_30_CNT_SOCIAL_CIRCLE (corr=0.8605)
cc_max_utilization	Redundant with cc_avg_utilization (corr=0.8502)
bureau_utilization_ratio	Range violation: min=-3112461135000.0, max=2250000000000.0
inst_max_days_late	Range violation: min=-156.0, max=2884.0

Columns recommended for removal before preprocessing and modeling, with reason for each recommendation.